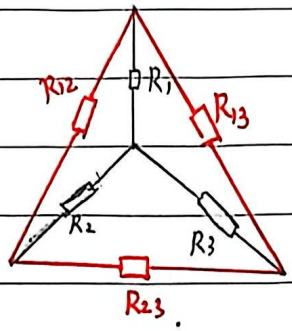
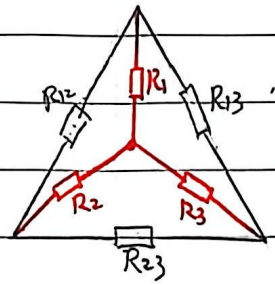
电阻的Y形联结与Δ形联结的等效变换.



$$\Rightarrow \begin{cases} R_{12} = \dfrac{R_1R_2 + R_1R_3 + R_2R_3}{R_3} \\ R_{13} = \dfrac{R_1R_2 + R_1R_3 + R_2R_3}{R_2} \\ R_{23} = \dfrac{R_1R_2 + R_1R_3 + R_2R_3}{R_1} \end{cases}$$



$$\Rightarrow \begin{cases} R_1 = \dfrac{R_{12}R_{13}}{R_{12}+R_{13}+R_{23}} \\ R_2 = \dfrac{R_{12}R_{23}}{R_{12}+R_{13}+R_{23}} \\ R_3 = \dfrac{R_{13}R_{23}}{R_{12}+R_{13}+R_{23}} \end{cases}$$

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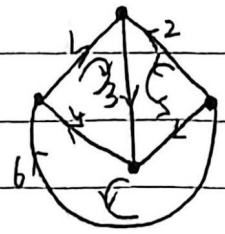
第三章 电阻电路的一般分析

小 n个结点的电路，独立的KCL方程为(n-1)个

n个结点、与b条支路的电路，树枝数为(n-1)，连枝数为b-(n-1)，即图的独立回路数

连支数 = 网孔数 = 独立回路数 = 独立KVL方程数



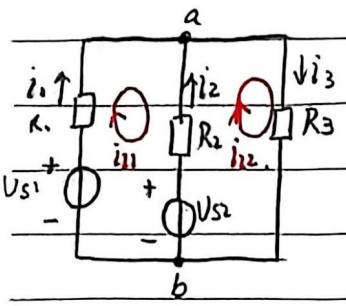
4个结点，6条支路，有树枝数为3，连支数为3，独立KVL方程数为3.

对网孔列KVL方程：

$$\begin{cases} u_1+u_3+u_4=0 \\ u_2+u_3-u_5=0 \\ u_4+u_5-u_6=0 \end{cases}$$

(2) 支路电流法.

(3) 网孔电流法.



$i_1=i_{l1}$; $i_2=i_{l2}-i_{l1}$; $i_3=i_{l2}$.

网孔1：$R_1 i_{l1}+R_2(i_{l1}-i_{l2})-U_{S1}+U_{S2}=0$

网孔2：$R_2(i_{l2}-i_{l1})+R_3 i_{l2}-U_{S2}=0$

整理得.

$$\begin{cases} (R_1+R_2)i_{l1}-R_2 i_{l2}=U_{S1}-U_{S2} \\ -R_2 i_{l1}+(R_2+R_3)i_{l2}=U_{S2} \end{cases}$$

⇓

一般方程形式：$\begin{cases} R_{11}i_{l1}+R_{12}i_{l2}=U_{Sl1} \\ R_{21}i_{l1}+R_{22}i_{l2}=U_{Sl2} \end{cases}$.

自电阻：R_{ii}；网孔i所有电阻之和；总为正.

互电阻：R_{ij}；网孔i与网孔j之间的互电阻.

U_{Sli}：网孔i中所有电压源电压代数和

~~当电压~~

对于具有 l 个网孔的电路，有：

$$\begin{cases} R_{11}i_{l1} + R_{12}i_{l2} + \cdots + R_{1l}i_{ll} = U_{sl1} \\ R_{21}i_{l1} + R_{22}i_{l2} + \cdots + R_{2l}i_{ll} = U_{sl2} \\ \cdots \\ R_{l1}i_{l1} + R_{l2}i_{l2} + \cdots + R_{ll}i_{ll} = U_{sll} \end{cases}$$

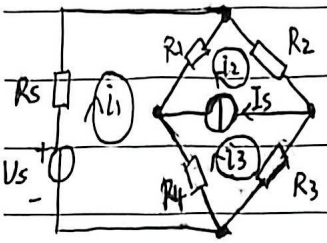
注意：R_{kk}：自电阻（总为正）

R_{jk}：互电阻 → $\begin{cases} +：流进互电阻的两个网孔电流方向相同 \\ -：\sim\sim\sim

(4) 回路电流法(网孔电流法的拓展)

求 I_S

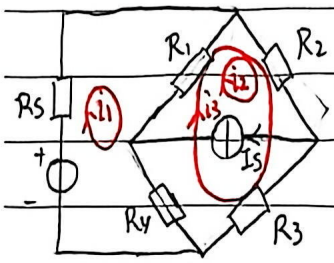
网孔法:



$$\begin{cases}(R_S+R_1+R_4)i_1-R_1i_2-R_4i_3=U_S\\ -R_1i_1+(R_1+R_2)i_2=U\\ -R_4i_1+(R_3+R_4)i_3=-U\end{cases}$$

增补 $I_S=i_2-i_3$.

回路电流法:

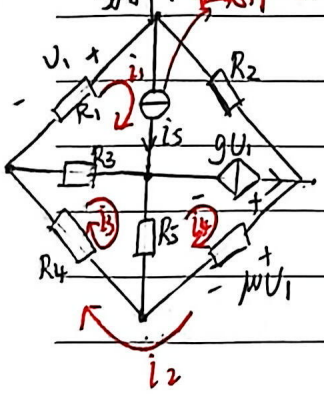


$$(R_S+R_1+R_4)i_1-R_1i_2-(R_1+R_4)i_3=U_S$$

~~R_1i_1~~ $i_2=i_S$.

$$-(R_1+R_4)i_1+(R_1+R_2)i_2+(R_1+R_2+R_3+R_4)i_3=0$$

分析: 受无伴电流源只有一个网孔经过

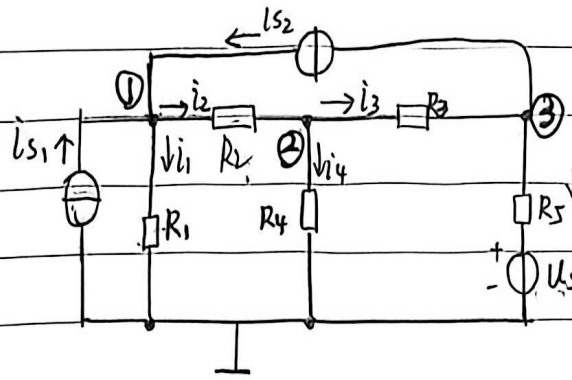


$$\begin{cases}i_S=i_1\\ R_1i_1+(R_1+R_2+R_4)i_2+R_4i_3=-\mu U_1\\ -R_3i_1+R_4i_2+(R_3+R_4+R_5)i_3-R_5i_4=0\\ \sout{i_4}\ i_4=gU_1\end{cases}$$

增补方程: $U_1=-R_1(i_1+i_2)$

④结点电压法（适用于结点较少的电路）

- 由与电流源串联电阻不计入方程
- 要将载流支路转化为伏安



对结点分析：

$$\begin{cases} i_1 - i_2 = i_{S1} + i_{S2} \\ -i_2 + i_4 + i_3 = 0 \\ -i_3 + i_5 = i_{S2} \end{cases}$$

整理得：

$$\begin{cases} \frac{U_{n1}}{R_1} + \frac{U_{n1}-U_{n2}}{R_2} = i_{S1} + i_{S2} \\ -\frac{U_{n1}-U_{n2}}{R_2} + \frac{U_{n2}-U_{n3}}{R_3} + \frac{U_{n2}}{R_4} = 0 \\ -\frac{U_{n2}-U_{n3}}{R_3} + \frac{U_{n3}-U_S}{R_5} = -i_{S2} \end{cases}$$

$$\begin{cases} \left(\frac{1}{R_1}+\frac{1}{R_2}\right)U_{n1} - \left(\frac{1}{R_2}\right)U_{n2} = i_{S1} + i_{S2} \\ -\frac{1}{R_2}U_{n1} + \left(\frac{1}{R_2}+\frac{1}{R_3}+\frac{1}{R_4}\right)U_{n2} - \frac{1}{R_3}U_{n3} = 0 \\ -\frac{1}{R_3}U_{n2} + \left(\frac{1}{R_3}+\frac{1}{R_5}\right)U_{n3} = -i_{S2} + \frac{U_S}{R_5} \end{cases}$$

$$\Downarrow$$

$$\begin{cases} G_{11}U_{n1} + G_{12}U_{n2} + G_{13}U_{n3} = i_{sn1} \\ G_{21}U_{n1} + G_{22}U_{n2} + G_{23}U_{n3} = i_{sn2} \\ G_{31}U_{n1} + G_{32}U_{n2} + G_{33}U_{n3} = i_{sn3} \end{cases}$$

$G_{11} = G_1 + G_2$　结点1的自导点

$G_{22} = G_2 + G_3 + G_4$　结点2的自导点

$G_{33} = G_3 + G_5$　结点3的自导点

结点的自导点等于接在该结点上所有支路的电导之和。

$G_{12} = G_{21} = -G_2$　结点1与结点2之间的互电导

$G_{23} = G_{32} = -G_3$　结点2与结点3之间的互电导。

互电导为接在结点与结点之间所有支路电导之和，总为负值

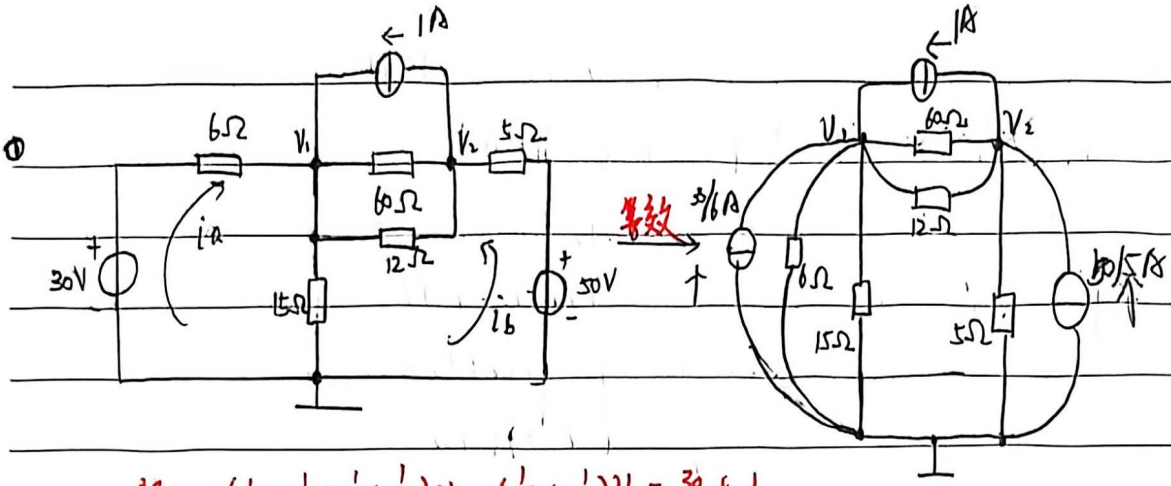
$i_{sn1} = i_{S1} + i_{S2}$　流入结点1的电流源电流的代数和

$i_{sn2} = 0$　结点2 〃

$i_{sn3} = -i_{S2} + \frac{U_S}{R_5}$　结点3 〃

流入结点取正号，流出取负号

①



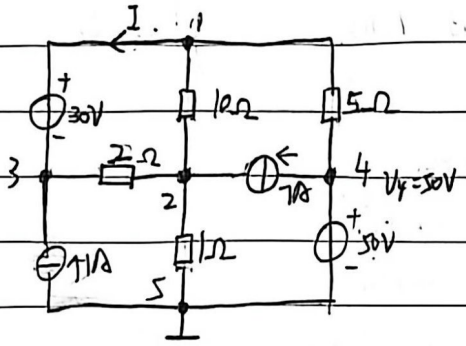
解: $\begin{cases}(\frac{1}{6}+\frac{1}{15}+\frac{1}{12}+\frac{1}{60})V_1-(\frac{1}{60}+\frac{1}{12})V_2=\frac{30}{6}+1\\ -(\frac{1}{60}+\frac{1}{12})V_1+(\frac{1}{60}+\frac{1}{12}+\frac{1}{5})V_2=-\frac{50}{5}-1\end{cases}$

$\therefore \begin{cases}V_1=30V\\ V_2=40V\end{cases}$

$V_{12}=V_1-V_2=-10V$

~~$i_a=\frac{30}{15}=2A$~~ $\begin{cases}6i_a=30+V_1\\ 5i_b=50+V_2\end{cases}$ ~~$i_a=0$~~ $i_b=2A$

② 独电压电源支路.[无伴电压源]



求 V_{12}.

解: $\begin{cases}(\frac{1}{10}+\frac{1}{5})V_1-\frac{1}{10}V_2-\frac{1}{5}V_4=-I\\ -\frac{1}{10}V_1+(\frac{1}{10}+\frac{1}{2}+1)V_2-\frac{1}{2}V_3=I\\ -\frac{1}{2}V_2+\frac{1}{2}V_3=II+1\\ V_4=50\\ V_1-V_3=30\end{cases}$

$V_1=40V$

$V_2=10V$

$V_{12}=V_1-V_2=30V$

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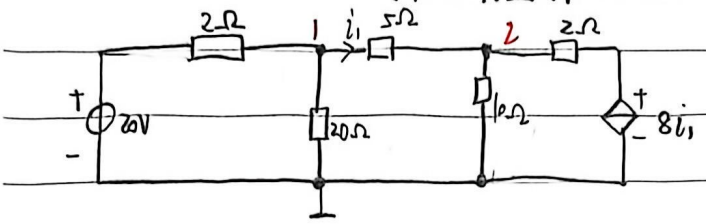
③含受控源电路的节点分析：{为把受控源作独立源处理 / 再将控制量用节点电压表示



用节点分析法确定2Ω电阻的功率.

解：$\begin{cases}(\frac{1}{2}+\frac{1}{20}+\frac{1}{5})V_1-\frac{1}{5}V_2=\frac{20}{2}\\ -\frac{1}{5}V_1+(\frac{1}{5}+\frac{1}{10}+\frac{1}{2})V_2=\frac{8i_1}{2}\\ i_1=\frac{V_1-V_2}{5}\end{cases}$

$\therefore V_1=16V \quad i_1=1.2A$

$V_2=10V \quad P_{5\Omega}=5i_1^2=7.2W.$

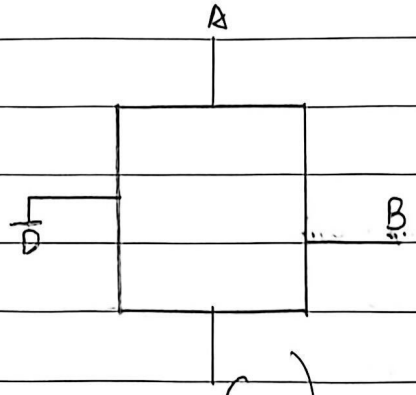
④ 4个节点电路，当$V_D=0$时，有

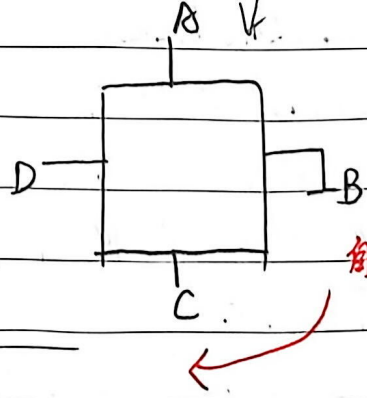
A: $3V_A-V_B=2$

B: $-V_A+5V_B-2V_C=0$

C: $-2V_B+3V_C=0$

改为V_B接地，修改方程.





自导=互导之和.

	A	B	C	D
A	3	1	0	2
B	1	5	2	2
C	0	2	3	1
D	2	2	1	5

A: $3V_A-2V_D=2$

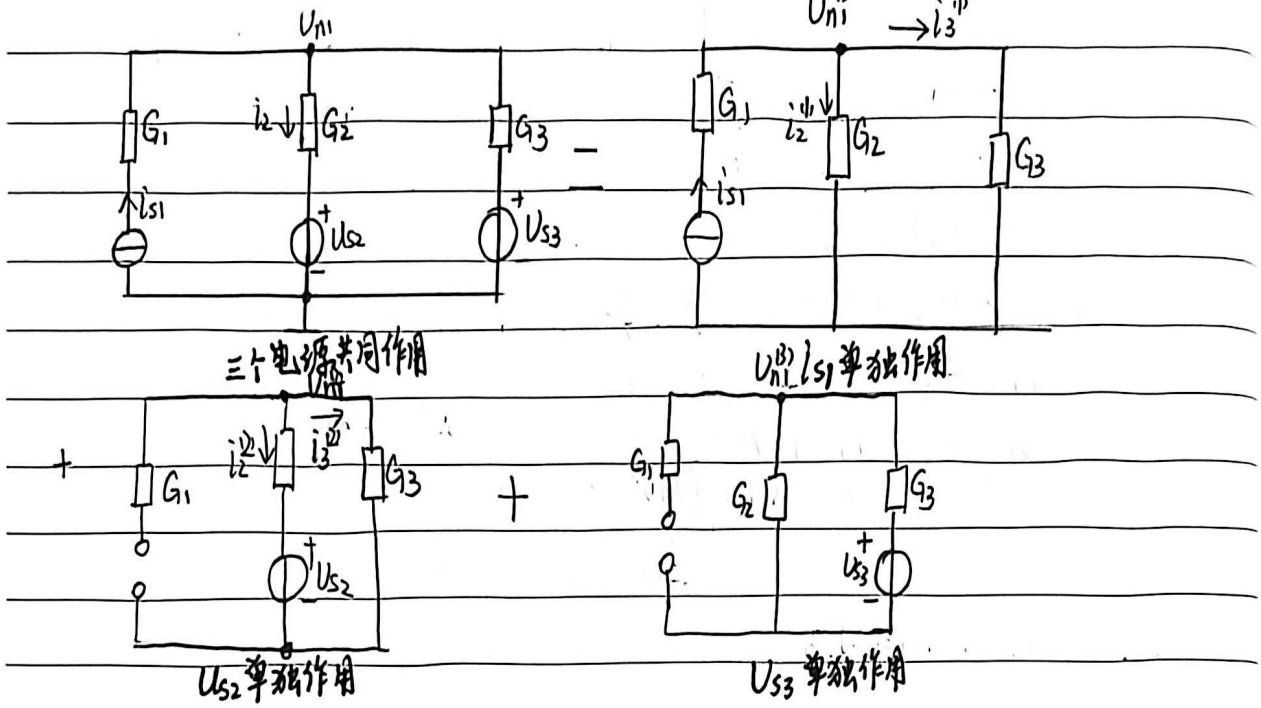
B: $V_B=0$

C: $3V_C-V_D=0$

D: $-2V_A-V_C+5V_D=-2$

[第四章　电路定理]

§ 4.1 叠加原理.



$$\begin{cases} U_{n1} = U_{n1}' + U_{n1}'' + U_{n1}''' \\ i_2 = i_2' + i_2'' + i_2''' \\ i_3 = i_3' + i_3'' + i_3''' \end{cases}$$

几点说明: 1. 叠加原理只适用于线性电路.

2. 二个电源作用, 其余电源为零.

电压源为零 — 短路.

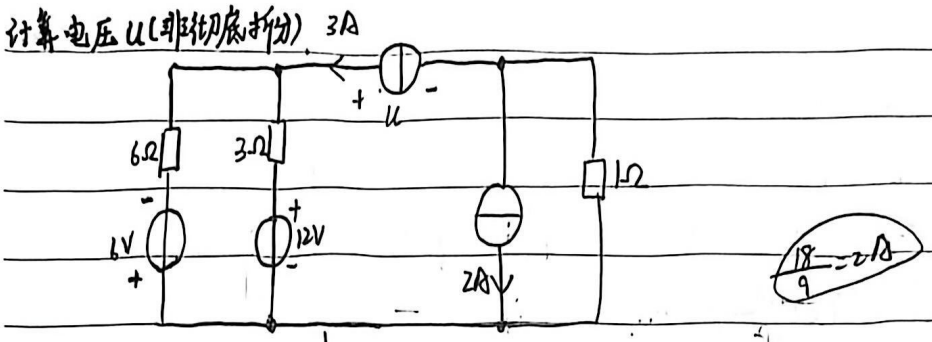
电流源为零 — 断路(开路).

3. 功率不能叠加.

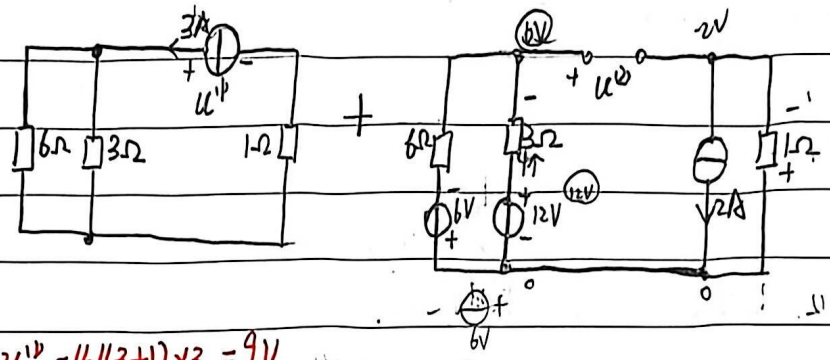
4. 以1叠加要注意各分量参考方向.

5. 含受控源(线性)电路亦可叠加, 但受控源均保留.

计算电压 u（非独底拆分）



$\frac{18}{9}=2A$



$u'=(6\|3+1)\times 3=9V$

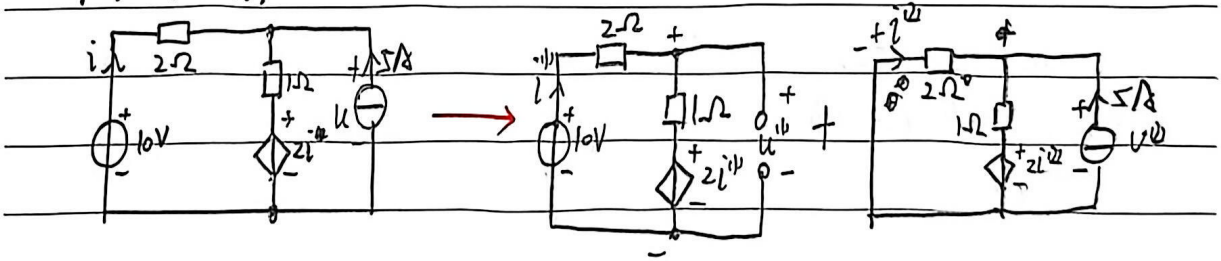
$u''=-\left(\frac{12+6}{6}\cdot 3\right)+12-[1\times(-2)]=8V$

$u=u'+u''=17V$

$2i''+(5+i'')\times 2+2i''=0$

$5i''=-15$

计算电压 u，电流 i.



解：$\frac{10-2i'}{3}=i' \Rightarrow i'=2A \quad u'=2\times 1+2\times 2=6V$

$2i''+(5+i'')\times 1+2i''=0 \Rightarrow i''=-1A \quad u''=-i''\times 2=2V$

$\therefore u=6+2=8V \quad i=2+(-1)=1A$

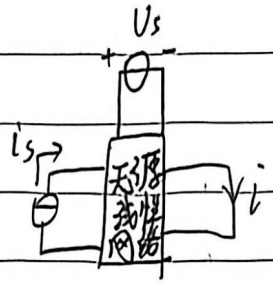
齐性原理.

封装好的电路如图，已知

当 $U_S=1V$, $i_S=1A$ 时，响应 $i=2A$

$U_S=1V$, $i_S=2A$ 时，响应 $i=1A$

求 $U_S=-3V$，$i_S=5A$ 时，响应 $i=?$

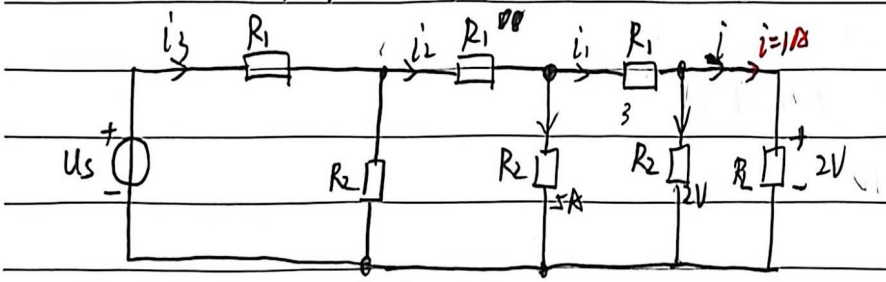


根据叠加原理：$i=K_1i_S+K_2U_S$.

$$\begin{cases} K_1+K_2=2 \\ 2K_1-K_2=1 \end{cases} \quad \begin{cases} K_1=1 \\ K_2=1 \end{cases}$$

$i=U_S+i_S=-3A+5A=2A$

例. $R_L=2\Omega$，$R_1=1\Omega$，$R_2=1\Omega$，$U_S=51V$，求电流 i.



采用倒推法

假设 $i=1A$，则 $U_{RL}=2V$

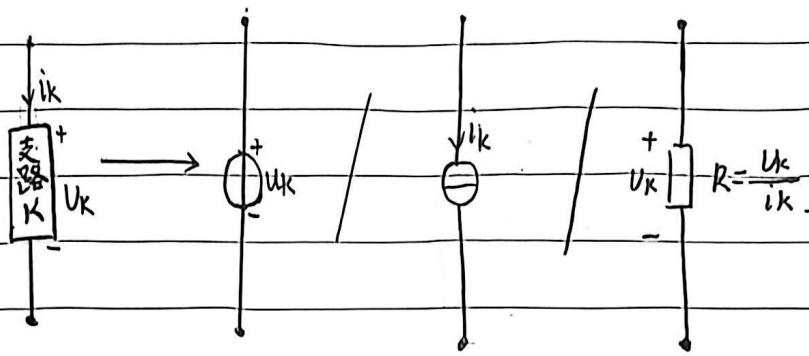
$i_1=3A$，$i_2=8A$，$i_3=21A$

$U_S=34V$

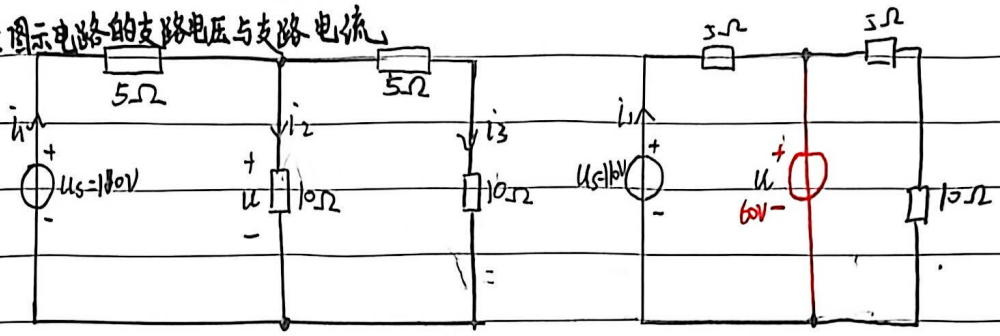
$\frac{34}{51}=\frac{1}{i}$

$i=1.5A$.

§ 4.2 替代定理.



① 求图示电路的支路电压与支路电流



解: $i_1=\frac{110}{[5+10//15]}=10A$ → 替代 → $i_1=\frac{110-60}{5}=10A$

$i_2=\frac{15}{10+15}i_1=6A$

$i_3=i_1-i_2=4A$.

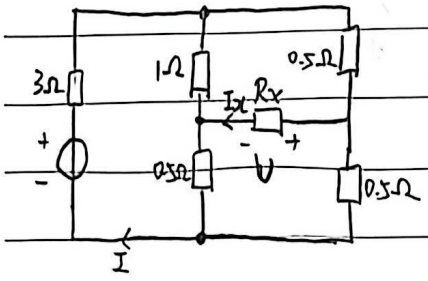
$u=10i_2=60V$

$i_3=60/(10+5)=4A$

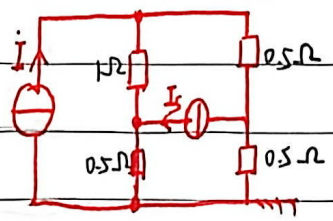
↓

替代后各支路电压与电流完全不变

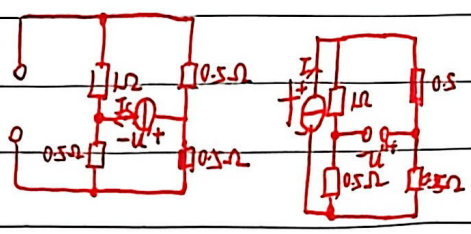
② 若使 $I_x=\frac{1}{8}I$, 求 R_x



解: 替代为



再用叠加原理.

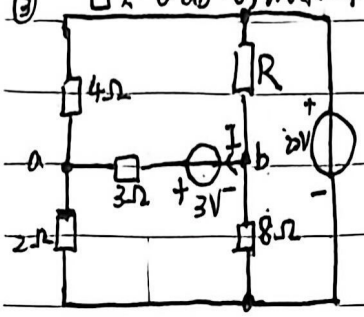


$u''+(0.5+0.5)\times\frac{1}{8}I\times\frac{1.5}{2.5}=-0.075I$

$u''=\frac{1}{2.5}\times I-\frac{1.5}{2.5}\times 0.5I=0.1I$

$u=u'+u''=0.025I$

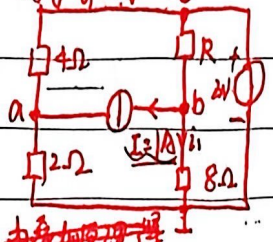
③ 已知 $U_{ab}=0$，求电阻 R.



解：对 a,b 支路分析.

$-3I+3=0 \Rightarrow I=1A$.

对可替代为：



由结点电压法可知：

$$\begin{cases}(\frac{1}{2}+\frac{1}{4})U_a-\frac{1}{4}U_c=I\\ U_c=20V\end{cases}$$

$\therefore U_a=8V=U_b$

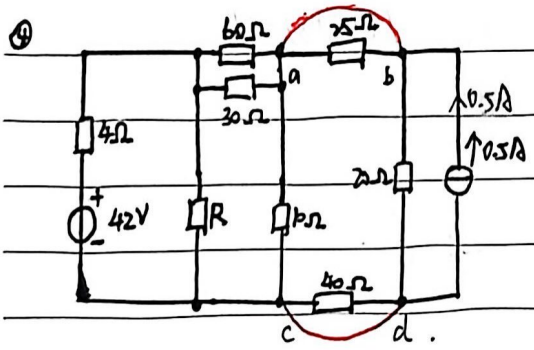
$I_R=I+i_1=1A+1A=2A$

$U_R=U_c-U_b=12V$

$\therefore R=\frac{U_R}{I_R}=6\Omega$

※：a,b 间电压为0，不能用电桥法 $\frac{2}{2}=\frac{R}{8}\Rightarrow R=16\Omega$，因为 ab 间有电流.

④

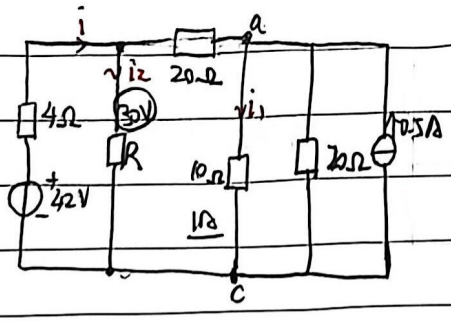


已知 $U_{ab}=0$，求电阻 R.

$U_{ab}=0$ → 用短路替代 或用断路替代.

用断路替代得 $U_{bd}=10V$，$U_{cd}=0$

用短路替代 ab,cd 得 $U_{ac}=U_{bd}=10V$



$U_{ac}=30V$，$i_1=\frac{U_{ac}}{10}=1A$

$U_R=i_1(20+10)=30V$.

$i=i_2+i_1$

$i=\frac{U-U_R}{4}=3A$.

$\therefore i_2=2A$.

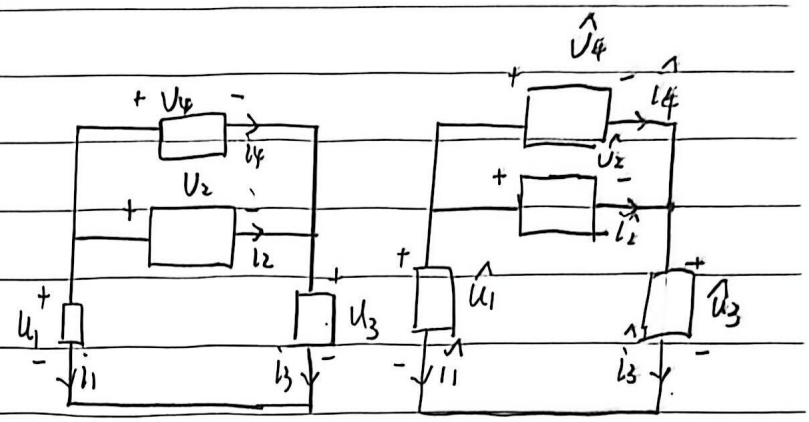
$R=\frac{U_R}{i_2}=15\Omega$

§4.3. 戴维宁定理与诺顿定理.

§4.5 特勒根定理.

① 形式一：功率守恒

② 形式二：似功率守恒



形式一：对于任意电路，假设其每个支路 u, i 关联方向，则 $\sum_{i=1} u_i i_i = 0$

即：$\begin{cases} u_1 i_1 + u_2 i_2 + u_3 i_3 + u_4 i_4 = 0 \\ \hat{u}_1 \hat{i}_1 + \hat{u}_2 \hat{i}_2 + \hat{u}_3 \hat{i}_3 + \hat{u}_4 \hat{i}_4 = 0 \end{cases}$

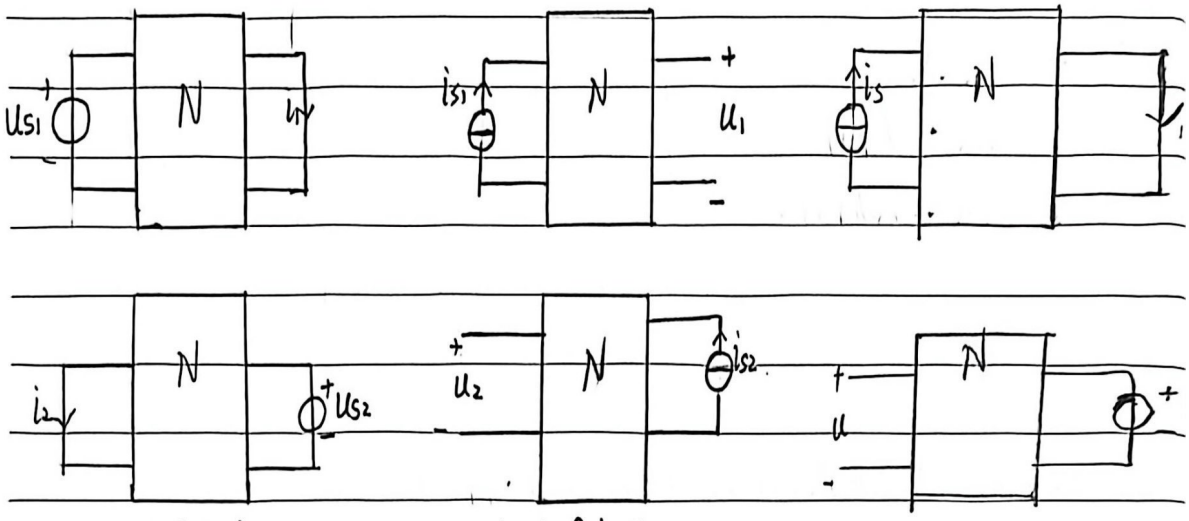
形式二：$\begin{cases} u_1 \hat{i}_1 + u_2 \hat{i}_2 + u_3 \hat{i}_3 + u_4 \hat{i}_4 = 0 \\ \hat{u}_1 i_1 + \hat{u}_2 i_2 + \hat{u}_3 i_3 + \hat{u}_4 i_4 = 0 \end{cases}$

注：① 特勒根与元件特性无关，是一种拓扑约束。

② u, i 都是瞬时值。

③ 方向。

互易定理.

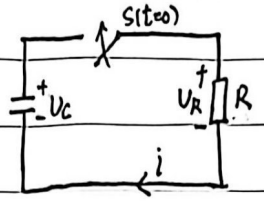


电压源&电流　　电流源&电压

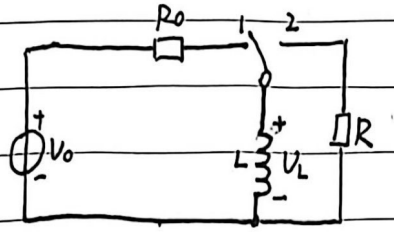
$U_{S1}i_2 = U_{S2}i_1$　　$U_2(-i_{S1}) = U_1(-i_{S2})$　　$U(-i_S) = U_S i$.

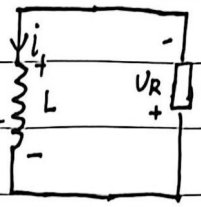
互易网N：①仅含线性R、L、C.　②L、C无储能.　③无独立源.

§ 7-2 一阶电路的零输入响应.

① 

$U_C = U_0 e^{-\frac{t}{RC}}$.

② 



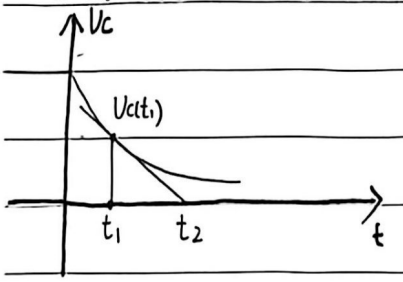
$i = i_0 e^{-\frac{t}{L/R}}$.

0输入

RC

$U_{C(0_-)} = U_0 = U_{(0_+)}$

$U_{C(t)} = U_0 e^{-\frac{t}{RC}}$ ($\tau = RC$).

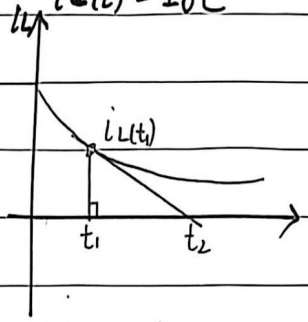


$\tau = t_2 - t_1$.

RL.

$i_{(0_-)} = I_0 = i_{(0_+)}$

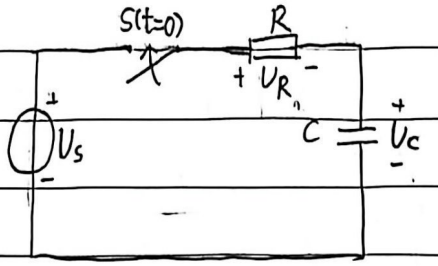
$i_{L(t)} = I_0 e^{-\frac{t}{L/R}}$ ($\tau = L/R$)



7-3 零状态响应.

① R-C.

$$U_C(t) = U_C(\infty)(1-e^{-\frac{t}{\tau}}) = U_C(\infty)(1-e^{-\frac{t}{RC}})$$



$U_{C(0-)} = U_{C(0+)} = 0$

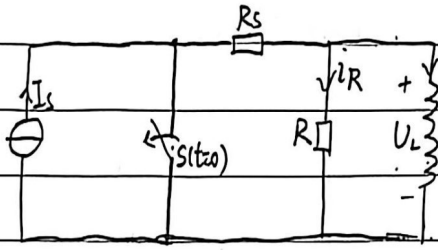
$t>0$

$U_{C(\infty)} = U_S$

$U_C(t) = U_S(1-e^{-\frac{t}{RC}})$

$$i = C\frac{du_C}{dt} = \frac{U_{(+\infty)}}{R}e^{-\frac{t}{RC}}$$

② R-L.



$i_{L(0-)} = i_{L(0+)} = 0$

当 $t>0$ 时.

$I_{L(\infty)} = I_S$

$I_L(t) = I_S(1-e^{-\frac{t}{\tau_R}})$

$$I_{L(t)} = I_{L(\infty)}(1-e^{-\frac{t}{\tau_R}})$$

$$U_L(t) = L\frac{di}{dt} = I_{L(\infty)}Re^{-\frac{t}{\tau_R}}$$

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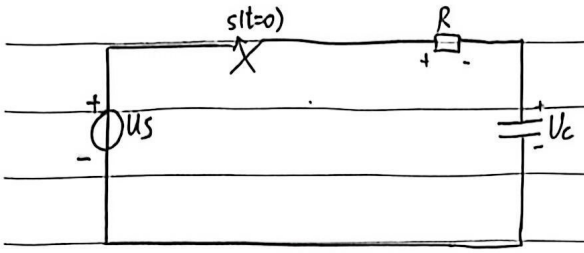
7-4 §. 一阶电路的全响应.

1) R-C.



$U_{C(0+)} = U_{C(0-)} = U_0$

$t>0$ 时

$iR + U_C = U_S$

$C\frac{dU}{dt} + U_C = U_S$

特解: $U_C' = U_S$

通解: $U_C'' = Ae^{-\frac{t}{RC}}$

$\therefore U_C = U_C' + U_C'' = U_S + Ae^{-\frac{t}{RC}}$

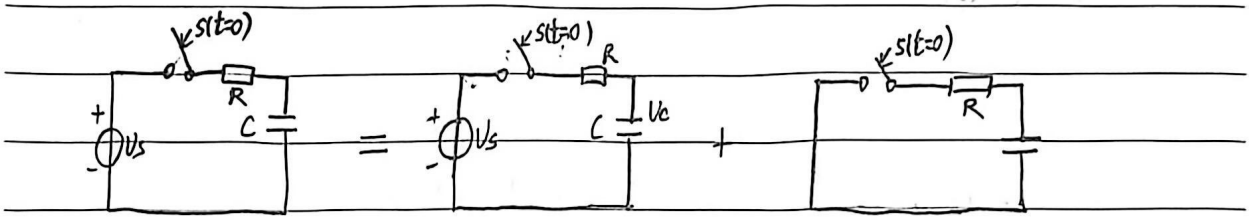
$U_{C(0+)} = U_S + A = U_0 \Rightarrow A = U_0 - U_S$

$\therefore U_C = U_S + Ae^{-\frac{t}{\tau}} = U_S + (U_0 - U_S)e^{-\frac{t}{\tau}} \quad t \geq 0$

$$U_{C(t)} = \underbrace{U_{(\infty)}}_{\text{稳态}} + (U_{0+} - U_{(\infty)})e^{-\frac{t}{\tau}}$$

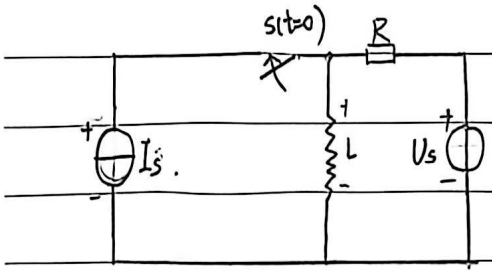
$$= U_{(\infty)}(1 - e^{-\frac{t}{\tau}}) + U_0 e^{-\frac{t}{\tau}} \cdot (t \geq 0)$$

零状态响应　　零输入响应



$U_{C(0-)} = U_0$　　$U_{C(0-)} = 0$　　$U_{C(0-)} = U_0$

2) R-L



$I_{L(0-)} = \frac{U_S}{R} = I_{L(0+)}$

$t>0$ 时.

$I_{L(\infty)} = I_S + \frac{U_S}{R}$.

$I_L = I_{L(\infty)} + (I_0 - I_\infty)e^{-\frac{t}{\tau}}$

$= I_0 e^{-\frac{t}{\tau}} + I_{L(\infty)}(1 - e^{-\frac{t}{\tau}})$

$$I_{L(t)} = I_\infty + (I_0 - I_\infty)e^{-\frac{t}{\tau}}$$

$$= I_\infty(1 - e^{-\frac{t}{\tau}}) + I_0 e^{-\frac{t}{\tau}}$$

零状态　　零输入

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§7.5 二阶电路的零输入响应.

§7.6 二阶电路的零状态响应与全响应.

① 欠阻尼：一对实部为负的共轭复根

临界阻尼：两个相等的负实根

过阻尼：两个不同负实根

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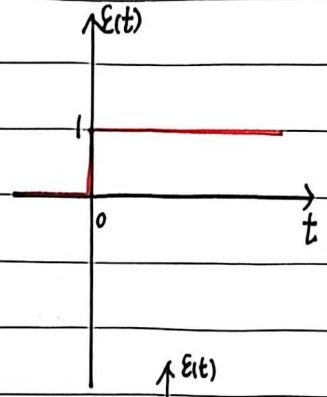
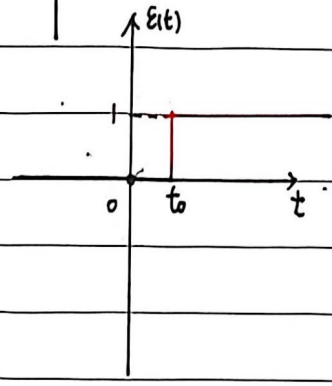
§7.7 一阶电路与二阶电路的阶跃响应.

1) 单位阶跃函数.



$$\varepsilon(t)=\begin{cases}0 & (t<0)\\ 1 & (t>0)\end{cases}$$

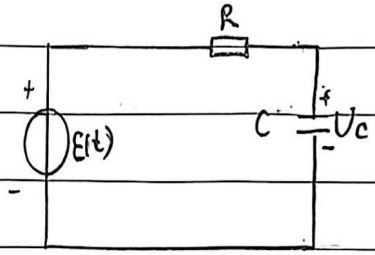


单位阶跃函数的延迟.

$$\varepsilon(t-t_0)=\begin{cases}0, & (t<t_0).\\ 1, & (t>t_0).\end{cases}$$

2) 一阶电路的阶跃响应:

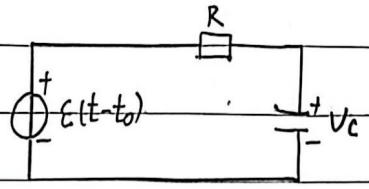
激励为单位阶跃函数时, 电路中产生的零状态响应.



$$\begin{cases}u_C(t)=\varepsilon(t)\left(1-e^{-\frac{t}{RC}}\right)\\ i(t)=\frac{1}{R}e^{-\frac{t}{RC}}\varepsilon(t).\end{cases}$$

$$u_C(0_-)=0$$

激励在 $t=t_0$ 时加入, 则响应从 $t=t_0$ 开始.

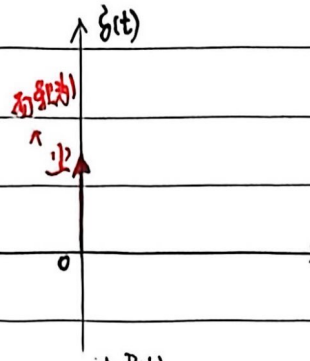


$$i_C=\frac{1}{R}e^{-\frac{t-t_0}{RC}}\varepsilon(t-t_0)$$

~~$i_C=\frac{1}{R}e^{-\frac{t}{RC}}\varepsilon(t-t_0)$~~

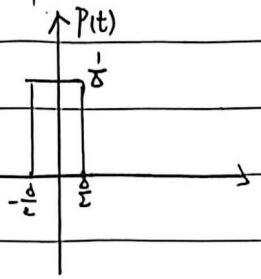
§ 7.8 一阶电路与二阶电路的冲激响应.

1) 单位冲激函数.



$$\begin{cases} \delta(t)=0 & (t\neq 0) \\ \int_{-\infty}^{\infty}\delta(t)dt=1. \end{cases}$$

~~$p(t)=\frac{1}{\Delta}[\varepsilon$~~

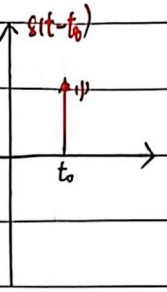


$$p(t)=\frac{1}{\Delta}\left[\varepsilon\left(t+\frac{\Delta}{2}\right)-\varepsilon\left(t-\frac{\Delta}{2}\right)\right]$$

$\Delta\to 0$ 时, $\frac{1}{\Delta}\to\infty$.

$\lim_{\Delta\to 0} p(t)=\delta(t)$.

2) 单位冲激函数的延迟.



$$\begin{cases} \delta(t-t_0)=0 & (t\neq t_0) \\ \int_{-\infty}^{\infty}\delta(t-t_0)dt=1=\int_{t_0-}^{t_0+}\delta(t-t_0)dt=1. \end{cases}$$

3) 单位冲激函数的性质.

① 冲激函数对时间的积分等于阶跃函数.

$$\int_{-\infty}^{t}\delta(t)dt=\begin{cases} 0, & t<0^- \\ 1, & t>0^+ \end{cases}=\varepsilon(t) \longleftrightarrow \frac{d\varepsilon(t)}{dt}=\delta(t)$$

4) 冲激函数的筛分性(取样性).

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt=f(0)\int_{-\infty}^{\infty}\delta(t)dt=f(0).$$

同理:

$$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt=f(t_0).$$

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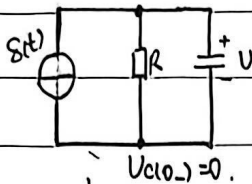
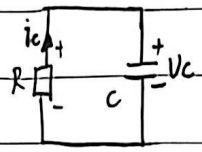
少一阶电路的冲激响应. → 激励为单位冲激函数时，电路中产生的零状态响应.

①



$U_{C(0_-)}=0$.



$t\in(0^-\to 0^+)$ 电容充电，有.

$$C\frac{du_c}{dt}+\frac{U_c}{R}=\delta(t)$$

$$\int_{0^-}^{0^+}C\frac{du_c}{dt}dt+\int_{0^-}^{0^+}\frac{U_c}{R}dt=\int_{0^-}^{0^+}\delta(t)dt=1$$

$$\Rightarrow C[U_{C(0_+)}-U_{C(0_-)}]=1.$$

$$\Rightarrow U_{C(0_+)}=\frac{1}{C}\neq U_{C(0_-)}.$$

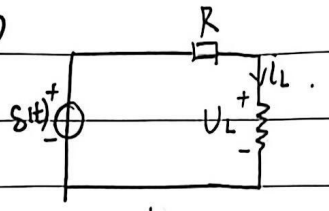
$t>0^+$ 时，为零输入响应.

$$\begin{cases}U_C=U_{C(0_+)}e^{-\frac{t}{\tau}}=\frac{1}{C}e^{-\frac{t}{RC}} & (t\geq 0^+)\\ i_C=-\frac{U_C}{R}=-\frac{1}{RC}e^{-\frac{t}{RC}} & (t>0^+)\end{cases}$$

综上：

$$\begin{cases}U_C=\frac{1}{C}e^{-\frac{t}{RC}}\varepsilon(t)\\ i_C=C\frac{du_c}{dt}=-\frac{1}{RC}e^{-\frac{t}{RC}}\varepsilon(t)+\underbrace{\varepsilon'(t)e^{-\frac{t}{RC}}}_{\delta(t)e^{-\frac{t}{RC}}=\delta(t)}\\ \quad=\delta(t)-\frac{1}{RC}e^{-\frac{t}{RC}}.\end{cases}$$

②



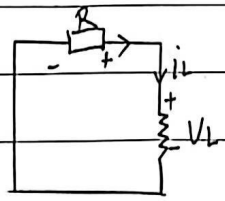
a: $t\in(0^-,0^+)$ 时，

$$Ri_L+U_L=\delta(t)\Rightarrow Ri_L+L\frac{di_L}{dt}=\delta(t)$$

$$\int_{0^-}^{0^+}Ri_Ldt+\int_{0^-}^{0^+}Ldi_L=\int_{0^-}^{0^+}\delta(t)dt=1.$$

$$L(i_{L(0^+)}-i_{L(0^-)})=1\Rightarrow i_{L(0^+)}=\frac{1}{L}\neq i_{L(0^-)}$$

b: $t>0^+$，为零输入响应.



$$\begin{cases}i_L=i_{L(0^+)}e^{-\frac{t}{\tau}}=\frac{1}{L}e^{-\frac{t}{\frac{L}{R}}}.\\ U_L=L\frac{di_L}{dt}=-\frac{R}{L}e^{-\frac{t}{\frac{L}{R}}}.\end{cases}$$

$$\Downarrow$$

综上：

$$\begin{cases}i_L=\frac{1}{L}e^{-\frac{t}{\tau}}\varepsilon(t)\\ U_L=\delta(t)-\frac{R}{L}e^{-\frac{t}{\tau}}\varepsilon(t).\end{cases}$$

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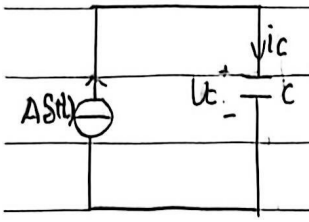
③电容电压或电感电流初值的跃变.

a:

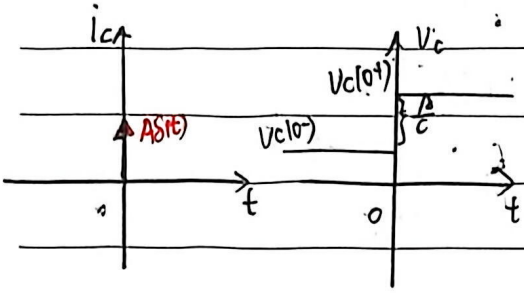


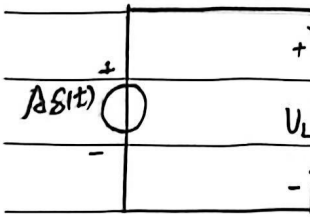
$$u_C(0^+) = u_C(0^-) + \frac{1}{C}\int_{0^-}^{0^+} i_C\,dt$$
$$= u_C(0^-) + \frac{A}{C}.$$

$$C\frac{du_C}{dt} = i_C \Rightarrow C\int_{-\infty}^{t} du_C = \int_{-\infty}^{0} i_C\,dt = \int_{-\infty}^{0^-} i_C\,dt + \int_{0^-}^{0^+} i_C\,dt + \int_{0^+}^{t} i_C\,dt$$

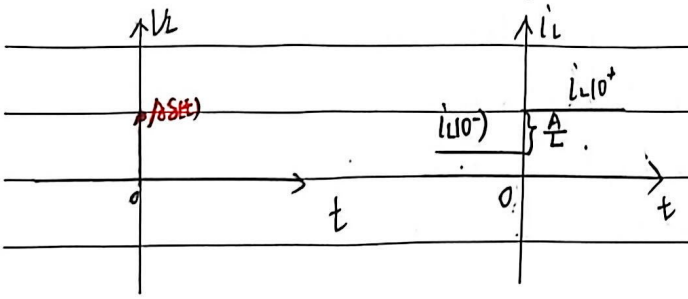
$$C\int_{-\infty}^{t} du_C = \int_{0^-}^{0^+} AS(t)\,dt$$

$$u_C(t^+) = u_C(t^-) + \frac{A}{C}$$





$$i_L(0^+) = i_L(0^-) + \frac{1}{L}\int_{0^-}^{0^+} u_L\,dt$$
$$= i_L(0^-) + \frac{A}{L}.$$



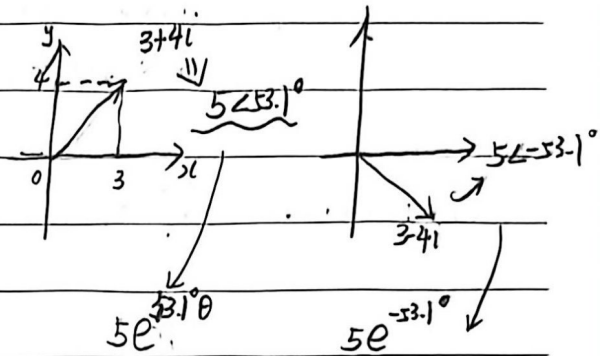
$\frac{10}{9}+(18\sqrt{6}+7)j$.

第八章. 相量法.

§ 8.1 复数

① 代数: $F=a+bi$.

复数 ② 极坐标: $F=|F|\angle\theta$.



③ 指数表示. $F=|F|e^{i\theta}$

(欧拉公式: $e^{i\theta}=\cos\theta+i\sin\theta$).

④ 共轭复数: $\overline{2+2i}=2-2i$.

(2) 复数运算.

① 加、减 一般用代数形式

② 乘、除 一般用极坐标形式.

$$\begin{cases} F_1\angle\theta_1\cdot F_2\angle\theta_2=F_1F_2\angle(\theta_1+\theta_2) \\ \frac{F_1\angle\theta_1}{F_2\angle\theta_2}=\frac{F_1}{F_2}\angle(\theta_1-\theta_2) \end{cases}$$

(3) 旋转因子

① $1\angle\theta$.

~~$(a+bi)\cdot 1$~~ $F\angle\alpha\cdot 1\angle\theta=F\angle(\alpha+\theta)$ 相当于把 $F\angle\alpha$ 逆时针旋转 θ.

② $\begin{cases} 1\angle 90°=i & \text{逆时针旋转}90° \\ 1\angle -90°=-i & -90° \\ 1\angle 180°=1\angle -180°=-1 & 180°/-180°. \end{cases}$

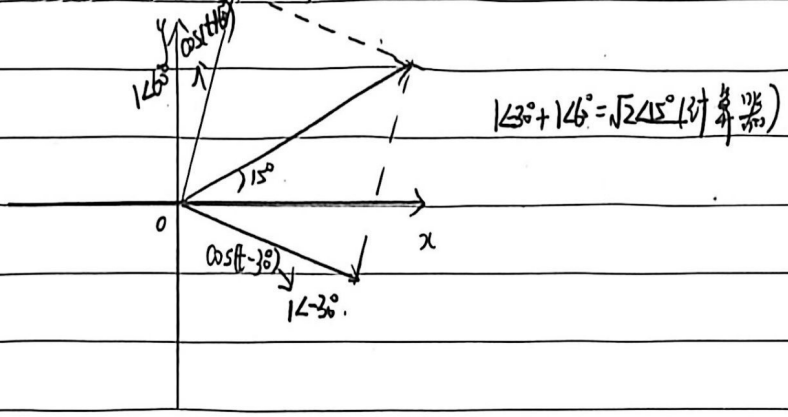
§8.2 正弦量.

1) ~~sin(ωt+φ)~~ $I_m\sin(\omega x+\varphi)$.

- I_m 幅值
- ω 角频率 rad/s. $\omega=\frac{2\pi}{T}=2\pi f$.
- φ 初相 $[-180°, 180°]$

2) 同频率正弦量的加减运算.

① $\cos(t-30°)+\cos(t+60°)=\cos(t+15°)$



② 适用条件: 同频率

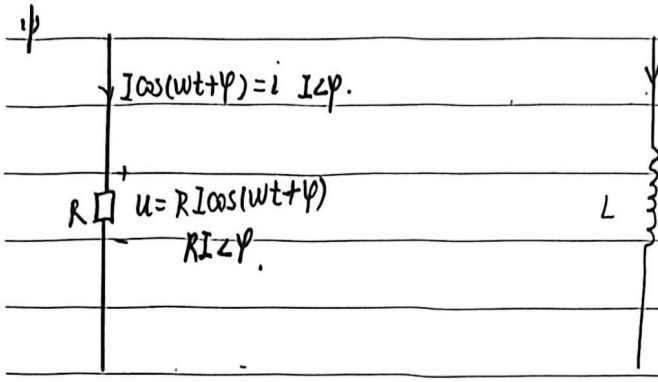
加减法,不适用于乘除.

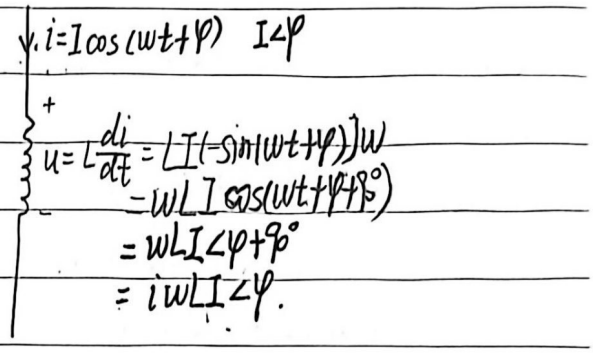
超前、滞后,[在$[-180°, 180°]$内讨论]

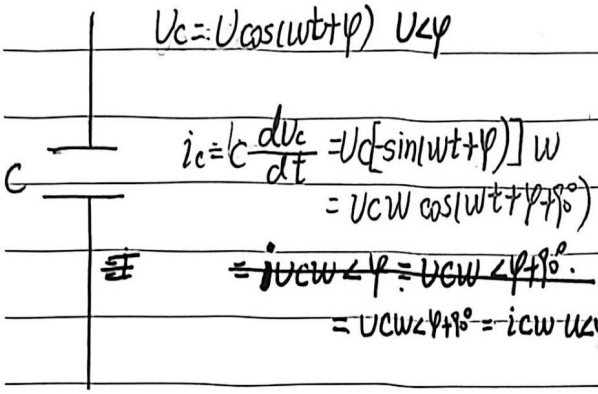
初相大超前

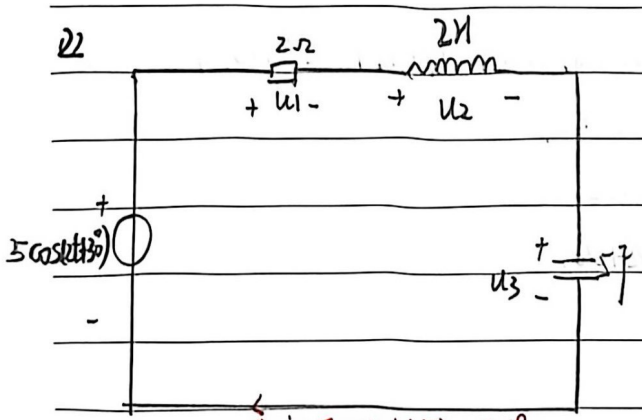
3) 有效值.

§. 8.3 相量法的基础









解：由KVL可知

$5\cos(2t+30°)=U_1+U_2+U_3$

$=2I\angle\varphi+j\omega LI\angle\varphi+\frac{I\angle\varphi}{j\omega C}$

$=(2+4j+\frac{1}{10\times j})I\angle\varphi$.

$=5\angle30°$.

$\therefore I\angle\varphi=\frac{5\angle30°}{2+4j+\frac{1}{10j}}=1.14\angle-32.9°$

$\Downarrow$

$i=1.14\cos(2t-32.9°)$.

相量法：激励为正弦函数

稳态时，各u,i同频率

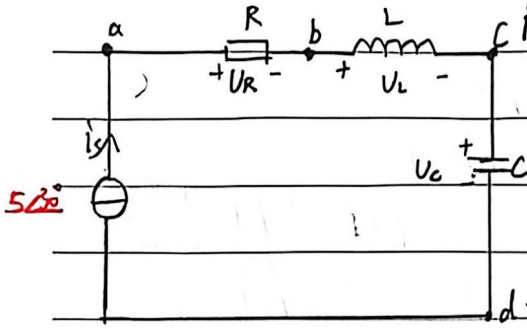
§8.3 电路定律的相量形式.

① 正弦激励下的电路：稳态　同频率(ω相同).

② 相量：
- 节点电流相量 约束
- 回路电压相量 约束.
- R、L、C、u、i 关系.

↓

- KCL　相量：$\dot{I}_1+\dot{I}_2+\cdots+\dot{I}_n=0$.
- KVL　相量：$\dot{U}_1+\dot{U}_2+\cdots+\dot{U}_n=0$
- VCR
  - R：$\dot{U}=R\dot{I}$
  - L：$\dot{U}=j\omega L\dot{I}$
  - C：$\dot{U}=\frac{1}{j\omega C}\dot{I}$



$i_S=5\sqrt{2}\cos(10^3t+30°)$A，$R=30\Omega$，$L=0.12$H，$C\approx 125\mu$F

求电压 U_{ad} 与 U_{bd}.

$U_{ad}=U_R+U_L+U_C$

$U_{bd}=U_L+U_C$.

$12.5\times10^{-3}\times10^{3}$

$\frac{1000}{125}=80$.

解：~~$\dot{U}_S=5\angle 0°$~~　$\dot{I}_S=5\angle 30°$

$\dot{U}_R=R\dot{I}_S=30\dot{I}_S$

$\dot{U}_L=j10^3\times0.12\dot{I}_S=120j\dot{I}_S$

$\dot{U}_C=\frac{1}{j\omega C}\dot{I}_S=\frac{1000}{12.5j}\times5\angle30°=\frac{400\angle30°}{j}=-400j\angle30°=400\angle-60°=-80j\dot{I}_S$

$\dot{U}_{ad}=\dot{U}_R+\dot{U}_L+\dot{U}_C=\dot{I}(30+120j-80j)=5\angle30°\times50\angle53.1°=250\angle83.1°$.

$\therefore u_{ad}=250\sqrt{2}\cos(10^3t+83.1°)$

$\dot{U}_{bd}=\dot{U}_L+\dot{U}_C=120j\dot{I}_S+(-80j)\dot{I}_S=40j\dot{I}_S=40\angle90°\cdot5\angle30°=200\angle120°$

$\therefore u_{bd}=200\sqrt{2}\cos(10^3t+120°)$

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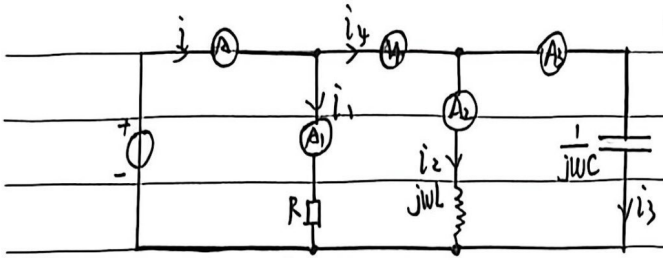


电表为交流电流表(测有效值),

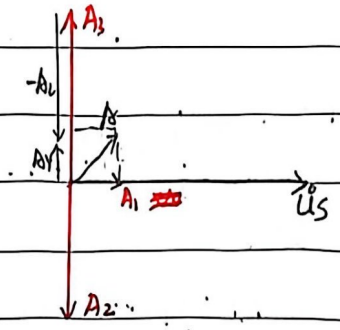
A_1读数为5A, A_2读数为20A; A_3读数为25A

求电流表A和A_4的读数.

解:(相量图法)

①指定参考相量.

指定 $\dot{U}_S$ 为~



$$\begin{cases} R & \dot{U}=R\dot{I} \\ L & \dot{U}=j\omega L\dot{I} \\ C & \dot{U}=\frac{1}{j\omega C}\dot{I} \end{cases}$$

$$\begin{cases} \dot{I}_4=\dot{I}_2+\dot{I}_3 \\ \dot{I}=\dot{I}_1+\dot{I}_4 \end{cases}$$

$$\begin{cases} I_{A4}=5A \\ I_A=5\sqrt{2}A \end{cases}$$

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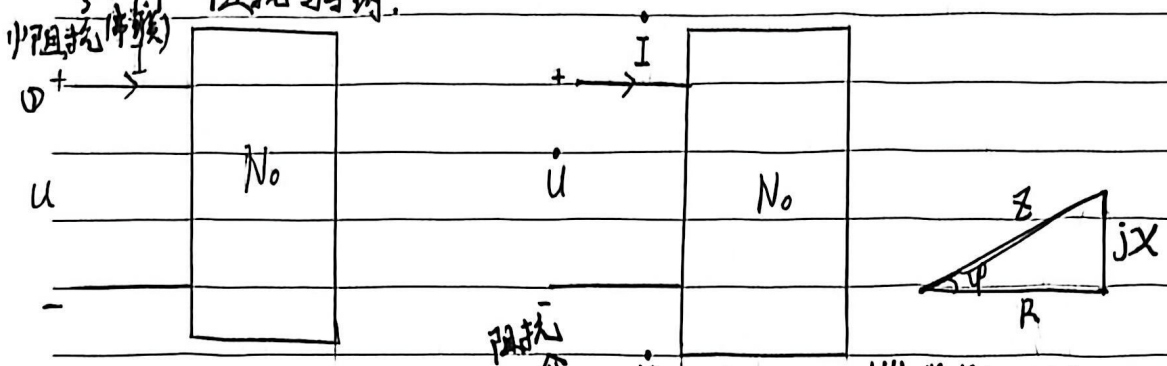
第九章　正弦稳态电路分析.

§9.1 阻抗与导纳.



纯阻抗(纯感)

$R_{eq}=\frac{U}{I}$ ∴

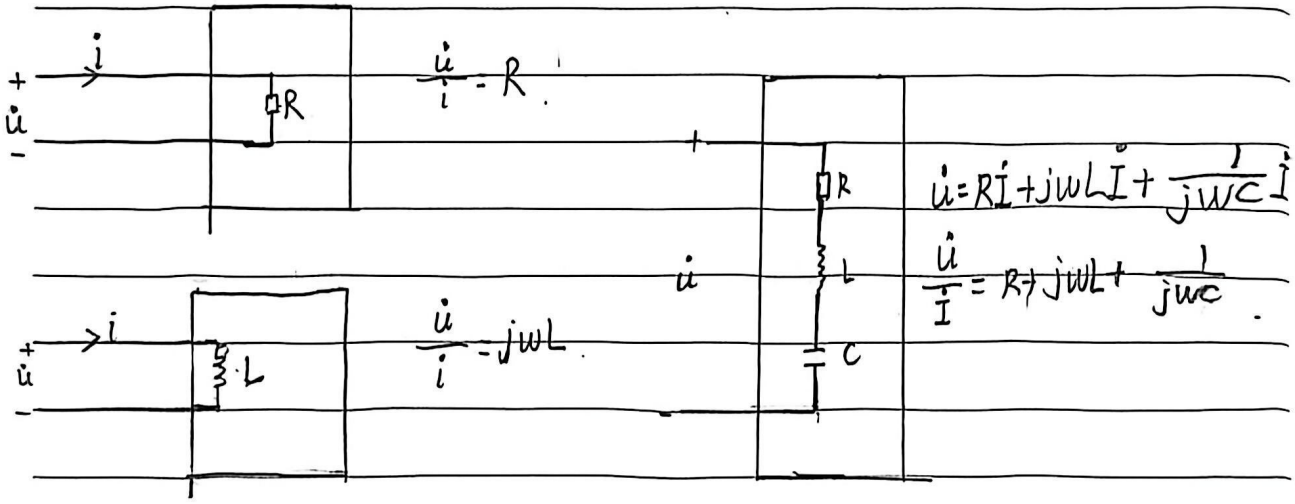
阻抗 $Z_{eq}=\frac{\dot U}{\dot I}=|Z_{eq}|\angle\varphi_z=\frac{U}{I}\angle(\varphi_u-\varphi_I)=R+jX$.

电阻　电抗

复数

$\varphi_u-\varphi_i>0$，$X>0$，呈电感性质

$\varphi_u-\varphi_i<0$，$X<0$，呈电容性质.

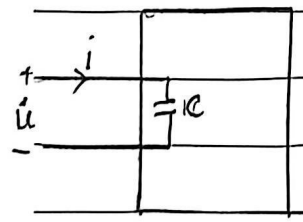


$\frac{\dot u}{\dot i}=R$.

$\frac{\dot u}{\dot i}=j\omega L$.

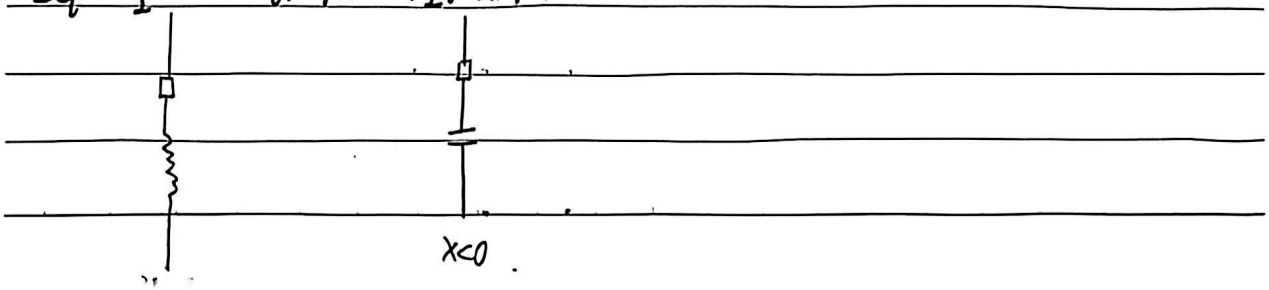
$\dot u=R\dot I+j\omega L\dot I+\frac{1}{j\omega C}\dot I$

$\frac{\dot U}{\dot I}=R+j\omega L+\frac{1}{j\omega C}$.

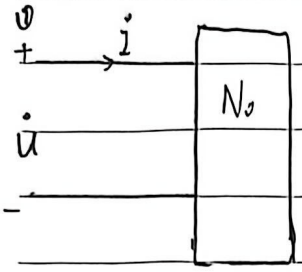


$\frac{\dot u}{\dot i}=\frac{1}{j\omega C}=-j\frac{1}{\omega C}$.

③ $Z_{eq}=\frac{\dot U}{\dot I}=|Z_{eq}|\angle\varphi_z=\frac{U}{I}\angle(\varphi_u-\varphi_i)=R+jX\Rightarrow\dot U=(R+jX)\dot I=R\dot I+jX\dot I$



② 导纳(并联)



$\frac{\dot{U}}{\dot{I}} = Z_{eq} = |Z_{eq}|\angle\varphi_Z = R+jX$

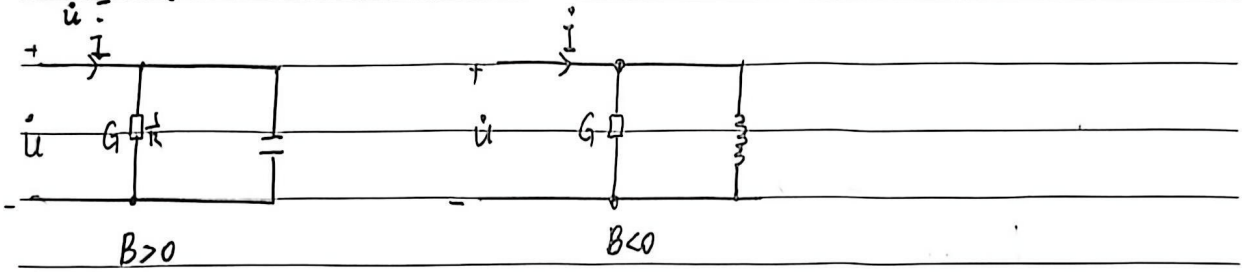
$\frac{\dot{I}}{\dot{U}} = \frac{1}{Z_{eq}} = \frac{1\angle 0}{|Z_{eq}|\angle\varphi_Z} = \frac{1}{|Z_{eq}|}\angle -\varphi_Z$ 引出导纳 $Y = G+jB$

电导 ↑ 电纳 ↑

$\varphi_Y = \varphi_i - \varphi_u$

$\begin{cases} \varphi_Y > 0, \text{电流领先, 电容性质, } B>0 \\ \varphi_Y < 0, \text{电压领先, 电感性质, } B<0 \end{cases}$

③ $\frac{\dot{I}}{\dot{U}} = G+jB \rightarrow \dot{I} = G\dot{U} + jB\dot{U}$

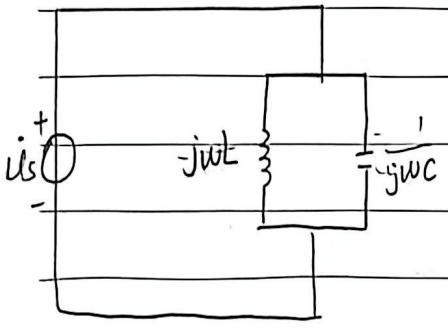


④ 串联谐振 & 并联谐振

当 $\dot{U}_S = \dot{I}(R+j\omega L+\frac{1}{j\omega C})$

当 $\omega L = \frac{1}{\omega C}$ 时, 电路可等效为只有 R.

称为串联谐振

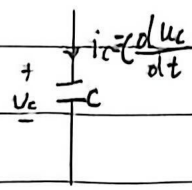


$Z_{eq} = \frac{-j\frac{1}{\omega C} \times j\omega L}{-j\frac{1}{\omega C} + j\omega L}$

当 $\omega L = \frac{1}{\omega C}$ 时, $Z_{eq} \rightarrow \infty$, 相当于开路, 则称为并联谐振.

第六章 储能元件

§6.1 储能元件

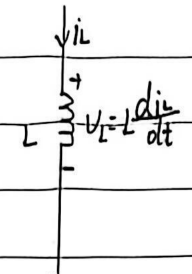


电容元件

单位：F、μF (10^{-6}F)

电压电流关系：$i_C=C\frac{du_C}{dt}$

存储能量：$W_C=\frac{1}{2}Cu_C^2$



电感元件

单位：H

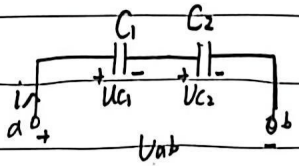
电压电流关系：$u_L=L\frac{di_L}{dt}$

存储能量：$W_L=\frac{1}{2}Li_L^2$

§6.2 电容电感的串并联

① 电容(与电阻相似)

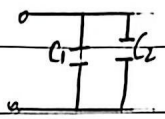
电容
- 电容的串联



$\frac{1}{C_{eq}}=\frac{1}{C_1}+\frac{1}{C_2}$

串联分压：$\begin{cases}u_{C1}=\frac{C_2}{C_1+C_2}u_{ab}\\ u_{C2}=\frac{C_1}{C_1+C_2}u_{ab}\end{cases}$

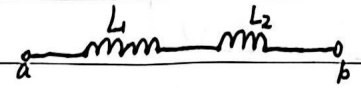
- 电容的并联



$C_{eq}=C_1+C_2$

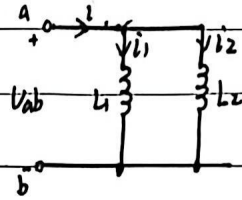
② 电感(与电阻相同)

电感
- 电感的串联



$L_{eq}=L_1+L_2$.

- 电感的并联



$\frac{1}{L_{eq}}=\frac{1}{L_1}+\frac{1}{L_2}$

并联分流：$\begin{cases}i_1=\frac{L_2}{L_1+L_2}i\\ i_2=\frac{L_1}{L_1+L_2}i\end{cases}$

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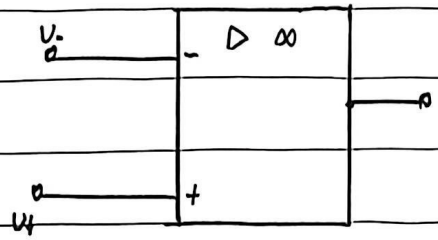
第五章 理想运算放大器



虚短：$u_+ = u_-$

虚断：输入端口电流为0

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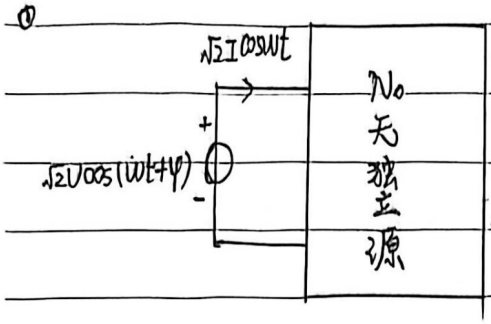
§9.4 正弦稳态电路的功率.

①



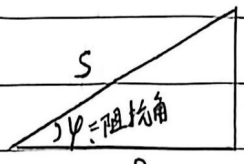
$p = ui = \sqrt{2}U\cos(\omega t+\varphi)\cdot\sqrt{2}I\cos\omega t$

$= UI\cos\varphi(1+\cos 2\omega t) - UI\sin\varphi\sin 2\omega t$

φ: u超前 I的角度.

① 平均功率(有功功率): $P = \frac{\int_0^T p\,dt}{T} = \frac{T\cdot UI\cos\varphi}{T} = UI\cos\varphi$ (U、I均为有效值). 单位: W.

② 无功功率: $Q = UI\sin\varphi$. 单位: var.
(能量交换最大瞬时功率)
·电容无功功率为负, 电感为正



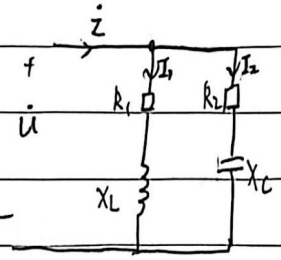
③ 视在功率: $S = UI$ V·A(单位)

②任一正弦稳态电路.

瞬时功率平衡. $\Sigma ui = 0$.

有功守恒: $\Sigma P = 0$

无功守恒. $\Sigma Q = 0$

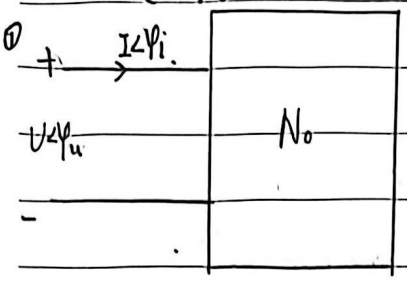


有功: $P = UI\cos\varphi = I_1^2R_1 + I_2^2R_2$

无功: $Q = UI\sin\varphi = I_1^2X_L - I_2^2X_C$

③ 功率因数: $\lambda(=\cos\varphi)$.

§. 9.5 复功率.

① 

复功率

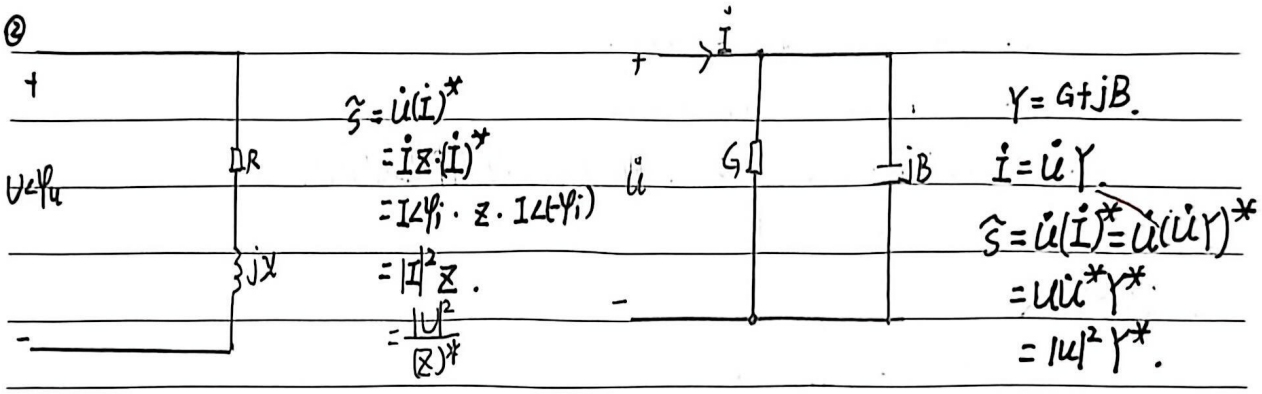
$\tilde{S} = \dot{U}(\dot{I})^* = U\angle\varphi_u \, I\angle-\varphi_i = UI\angle(\varphi_u-\varphi_i) = UI\cos\varphi + jUI\sin\varphi$. [U,I功有效]

视在功率. 有功功率 无功功率

eg: $\dot{U} = 6\angle 30°$, $\dot{I} = 2\angle -30°$

$\dot{U}(\dot{I})^* = 6\angle 30° \cdot 2\angle 30° = 12\angle 60° = 6 + 6\sqrt{3}i$

视在 有功 无功

② 

$\tilde{S} = \dot{U}(\dot{I})^*$

$= \dot{I}Z(\dot{I})^*$

$= I\angle\varphi_i \cdot Z \cdot I\angle-\varphi_i)$

$= |I|^2 Z$.

$= \frac{|U|^2}{(Z)^*}$

$Y = G + jB$.

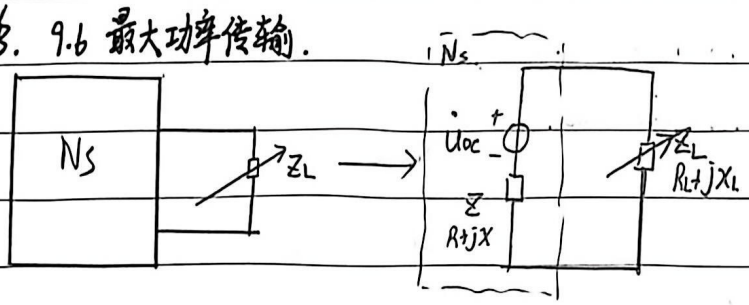
$\dot{I} = \dot{U}Y$.

$\tilde{S} = \dot{U}(\dot{I})^* = \dot{U}(\dot{U}Y)^*$

$= U\dot{U}^*Y^*$.

$= |U|^2 Y^*$.

§. 9.6 最大功率传输.



Ns不变; Z_L可变; 当五?, Z_L可获得最大有功功率?

① Z_L可任意变化, R_L, X_L都可随意变。 当 $Z_L = (Z)^*$, Z_L可获最大有功: $P_{max} = \frac{(U_{OC})^2}{4R}$ (平均有功)

(共轭匹配)

② $|Z_L|$可变, $Z_L = |Z_L|\angle\varphi$; φ固定。当 $|Z_L| = |Z|$ 时, 可获最大功率。

(若模匹配).

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第十章 含有耦合电感的电路

§10.1 互感：

① 磁链：$\psi = N\Phi$（匝数）

② 空心线圈，ψ 与 i 成正比。当只有1个线圈时，$\psi_1 = \psi_{11} = L_1 i_1$，$L$ 为自感系数，单位为亨(H)。

ψ_{ij} 下标 i：所在线圈；j：产生磁链的线圈。

③ 两个线圈都有电流时，每一线圈的磁链为自感磁通链与互感磁通链的代数和：

$$\begin{cases} \psi_1 = \psi_{11} \pm \psi_{12} = L_1 i_1 \pm M_{12} i_2 \\ \psi_2 = \psi_{21} + \psi_{22} = \pm M_{21} i_1 + L_2 i_2 \end{cases}$$

式中 M_{12}、M_{21} 为互感系数，单位为亨。（$M_{12} = M_{21}$ 与线圈中电流无关）

④ 耦合系数

$$K \xrightarrow{def} \frac{M}{\sqrt{L_1 L_2}} \leq 1$$

用耦合系数 K 表示两个线圈磁耦合的紧密程度。

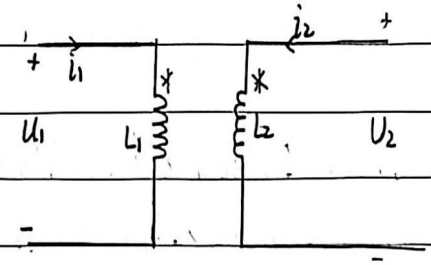
$\begin{cases} K=1 \text{ 称为全耦合：漏磁 } \Phi_{S1} = \Phi_{S2} = 0 \\ \text{满足：} \quad \Phi_{11} = \Phi_{21},\ \Phi_{22} = \Phi_{12} \end{cases}$

⑤ 耦合电感上电压、电流关系

$$u_1 = u_{11} \pm u_{12} = L_1 \frac{di_1}{dt} \pm M \frac{di_2}{dt}$$

（自感电压，互感电压）

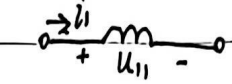
$$u_2 = \pm u_{21} + u_{22} = \pm M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$



①互感线圈的同名端.

A: 对于自感电压，当 u_1 i取关联参考方向，u_1 i与 Φ 符合右手螺旋定则，

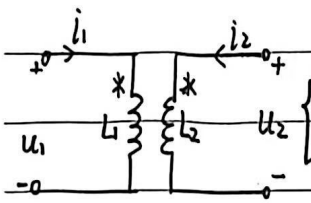
$$u_{11}=\frac{d\psi_{11}}{dt}=N_1\frac{d\Phi_{11}}{dt}=L_1\frac{di_1}{dt}$$



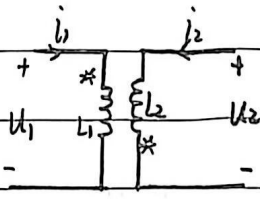
B: 同名端即当两个电流分别从两个线圈的对应端子同时流入或流出，若所产生的磁通互相加强时，则这两个对应端子称为两互感线圈的同名端.

也当随时间增大的时变电流从一线圈的一端流入时，将会引起另一线圈相应的同名端电位上升.

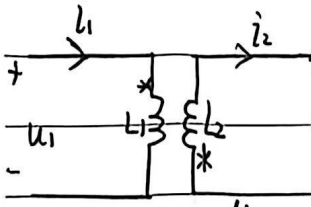
C:



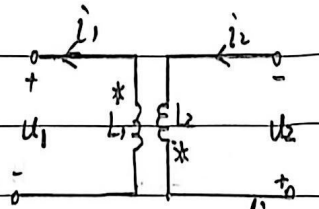
$$\begin{cases}u_1=L_1\frac{di_1}{dt}+M\frac{di_2}{dt}\\ u_2=M\frac{di_1}{dt}+L_2\frac{di_2}{dt}\end{cases}$$



$$\begin{cases}u_1=L_1\frac{di_1}{dt}-M\frac{di_2}{dt}\\ u_2=-M\frac{di_1}{dt}+L_2\frac{di_2}{dt}\end{cases}$$



$$\begin{cases}u_1=L_1\frac{di_1}{dt}+M\frac{di_2}{dt}\\ u_2=-M\frac{di_1}{dt}-L_2\frac{di_2}{dt}\end{cases}$$



$$\begin{cases}u_1=-L_1\frac{di_1}{dt}-M\frac{di_2}{dt}\\ u_2=-M\frac{di_1}{dt}-L_2\frac{di_2}{dt}\end{cases}$$

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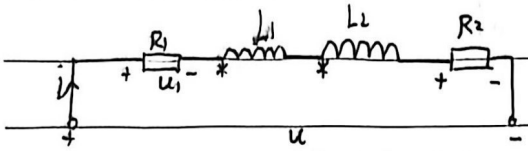
10.2 §. 含有耦合电感电路的计算.

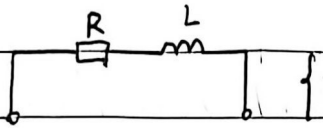
(1) 耦合电感的串联.

① 同向(顺接)串联.



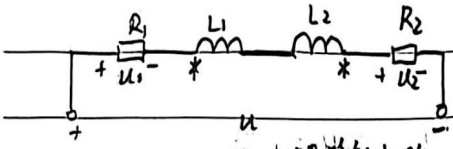
$$u = R_1 i + L_1\frac{di}{dt} + M\frac{di}{dt} + M\frac{di}{dt} + R_2 i$$
$$= (R_1+R_2)i + \frac{di}{dt}(L_1+L_2+2M)$$

⇓去耦等效电路



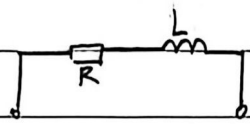
$$\begin{cases} R = R_1+R_2 \\ L = L_1+L_2+2M \end{cases}$$

② 反向(反接串联)



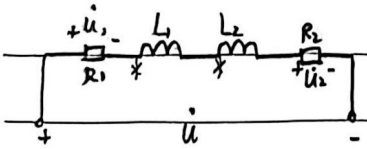
$$u = R_1 i + L_1\frac{di}{dt} - M\frac{di}{dt} + L_2\frac{di}{dt} - M\frac{di}{dt} + R_2 i$$
$$= i(R_1+R_2) + \frac{di}{dt}(L_1+L_2-2M)$$

⇓去耦等效电路

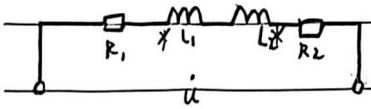


$$\begin{cases} R = R_1+R_2 \\ L = (L_1+L_2-2M) \quad (L_1+L_2-2M\geq 0 \Rightarrow M\leq\frac{1}{2}(L_1+L_2)) \end{cases}$$

在正弦激励下



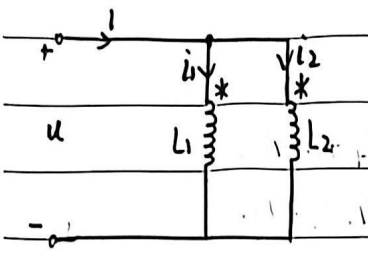
$$\dot{U} = \dot{I}(R_1+R_2) + j\omega(L_1+L_2+2M)\dot{I}$$



$$\dot{U} = \dot{I}(R_1+R_2) + j\omega(L_1+L_2-2M)\dot{I}$$

2. 含耦合电感的并联

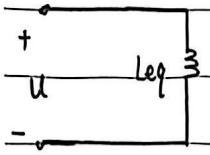
① 同侧(同名端)并联



$$\begin{cases} u = L_1\dfrac{di_1}{dt} + M\dfrac{di_2}{dt} \\ u = L_2\dfrac{di_2}{dt} + M\dfrac{di_1}{dt} \\ i = i_1 + i_2 \end{cases}$$

⇓去耦等效电路

$$u = \frac{(L_1L_2 - M^2)}{L_1 + L_2 - 2M}\cdot\frac{di}{dt}$$



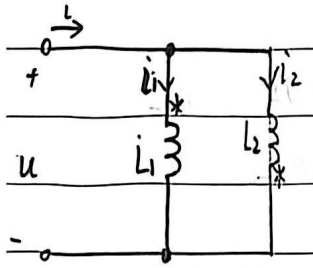
$$L_{eq} = \frac{L_1L_2 - M^2}{L_1 + L_2 - 2M} \geqslant 0$$

若全耦合：$L_1L_2 = M^2$

a：当 $L_1 \neq L_2$ 时，$L_{eq} = 0$（短路）

b：当 $L_1 = L_2 = L$ 时，$L_{eq} = L = \frac{L^2 - M}{2(L-M)} = \frac{L+M}{2} = L$（相当于导线加粗，电感不变）

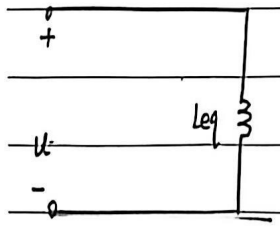
② 异侧(反接)并联



$$\begin{cases} u = L_1\dfrac{di_1}{dt} - M\dfrac{di_2}{dt} \\ u = L_2\dfrac{di_2}{dt} - M\dfrac{di_1}{dt} \\ i = i_1 + i_2 \end{cases}$$

⇓去耦等效电路

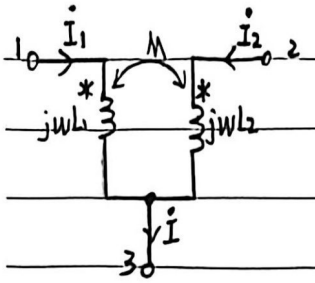
$$u = \frac{L_1L_2 - M^2}{L_1 + L_2 + 2M}\frac{di}{dt}$$



$$L_{eq} = \frac{L_1L_2 - M^2}{L_1 + L_2 + 2M}$$

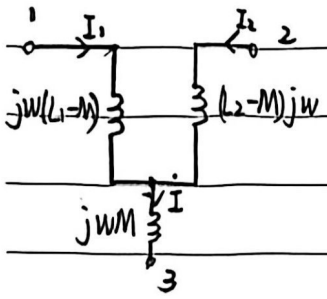
3) 耦合电感的T型等效.

① 同名端为共端的T型去耦等效.

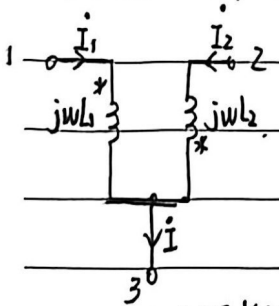


$$\begin{cases} \dot{U}_{13} = jwL_1\dot{I}_1 + jwM\dot{I}_2 = jw(L_1-M)\dot{I}_1 + jwM\dot{I} \\ \dot{U}_{23} = jwL_2\dot{I}_2 + jwM\dot{I}_1 = jw(L_2-M)\dot{I}_2 + jwM\dot{I} \\ \dot{I} = \dot{I}_1 + \dot{I}_2 \end{cases}$$

↓去耦等效电路.



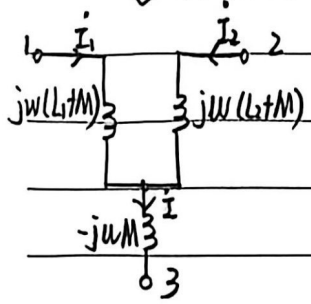
② 异名端为去端的T型去耦等效.



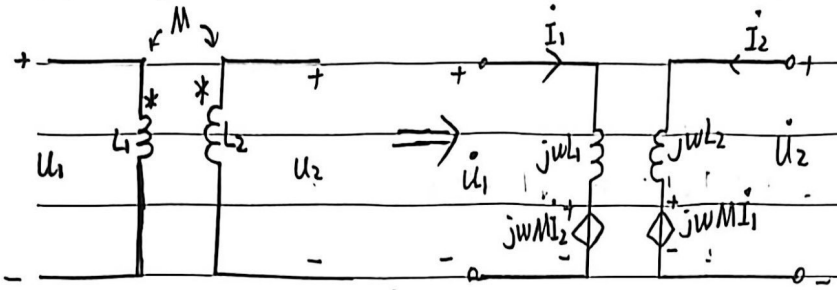
~~$\dot{U}_{13} = jwL_1\dot{I}_1$~~

$$\begin{cases} \dot{U}_{13} = jwL_1\dot{I}_1 - jwM\dot{I}_2 = jw(L_1+M)\dot{I}_1 - jwM\dot{I} \\ \dot{U}_{23} = jwL_2\dot{I}_2 - jwM\dot{I}_1 = jw(L_2+M)\dot{I}_2 - jwM\dot{I} \\ \dot{I} = \dot{I}_1 + \dot{I}_2 \end{cases}$$

↓去耦等效电路.

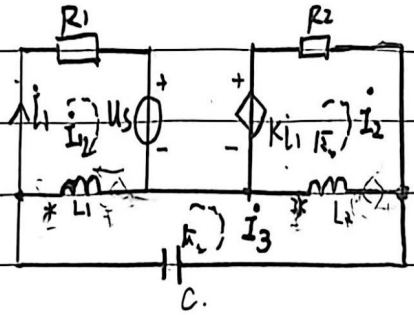


含受控源等效电路.



$$\begin{cases} \dot{U}_1 = jwL_1\dot{I}_1 + jwM\dot{I}_2 \\ \dot{U}_2 = jwM\dot{I}_1 + jwL_2\dot{I}_2 \end{cases}$$

(例)



用回路电流法分析电路.

$$\begin{cases} (R_1+jwL_1)\dot{I}_1 - jwL_1\dot{I}_3 + \cancel{jwM\dot{I}_3} - \cancel{jwM\dot{I}_2} = -\dot{U}_S - jwM(\dot{I}_2-\dot{I}_3) \\ (R_2+jwL_2)\dot{I}_2 - jwL_2\dot{I}_3 = K\dot{I}_1 - jwM(\dot{I}_1-\dot{I}_3) \\ (jwL_1+jwL_2-j\frac{1}{wc})\dot{I}_3 - jwL_1\dot{I}_1 - jwL_2\dot{I}_2 = -jwM(\dot{I}_3-\dot{I}_1) \\ \quad -jwM(\dot{I}_3-\dot{I}_2) \end{cases}$$

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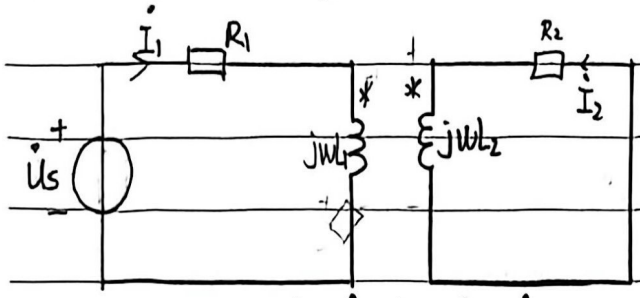
§10.3 耦合电感的功率



$$\begin{cases}(R_1+j\omega L_1)\dot{I}_1+j\omega M\dot{I}_2=\dot{U}_S\\(R_2+j\omega L_2)\dot{I}_2+j\omega M\dot{I}_1=0\end{cases}$$

$$\overline{S_1}=\dot{U}_S(\dot{I}_1)^*=(R_1+j\omega L_1)I_1^2+j\omega M\dot{I}_2(\dot{I}_1)^*$$

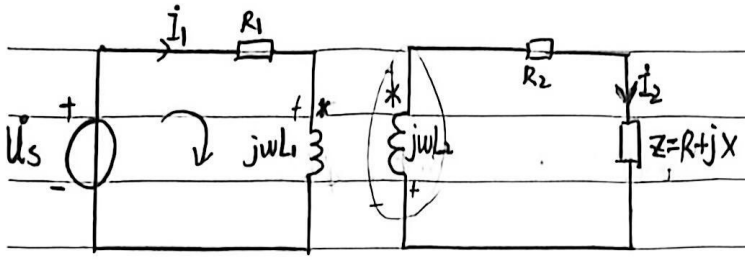
$$\overline{S_2}=0\cdot(\dot{I}_2)^*=0=(R_2+j\omega L_2)I_2^2+j\omega M\dot{I}_1(\dot{I}_2)^*$$

$j\omega M\dot{I}_2(\dot{I}_1)^*$ 线圈1中互感电压耦合的复功率.

$j\omega M\dot{I}_1(\dot{I}_2)^*$ 线圈2中互感电压耦合的复功率

(注)：两个互感电压耦合的复功率虚部同号，而实部异号

§10.4 变压器



$$\begin{cases}(R_1+j\omega L_1)\dot{I}_1-j\omega M\dot{I}_2=\dot{U}_S \\ (j\omega L_2+jX+R+R_2)\dot{I}_2-j\omega M\dot{I}_1=0\end{cases}$$

令 $Z_{11}=R_1+j\omega L_1$，$Z_{22}=(R+R_2)+j(\omega L_2+X)$.

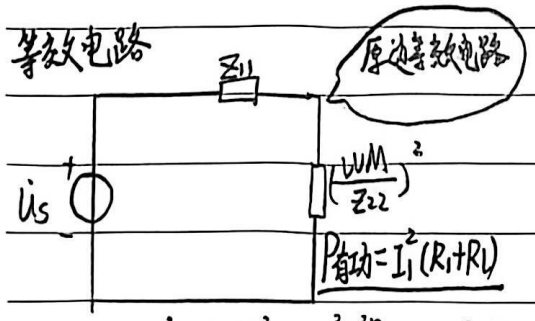
则 $\begin{cases}Z_{11}\dot{I}_1-j\omega M\dot{I}_2=\dot{U}_S \\ Z_{22}\dot{I}_2-j\omega M\dot{I}_1=0\end{cases}$

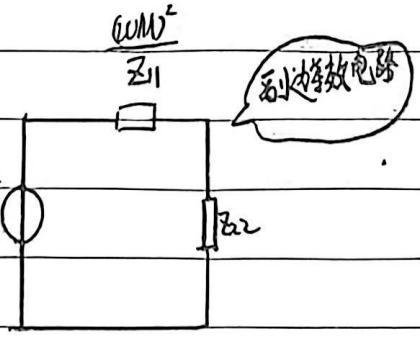
$$\dot{I}_1=\frac{\dot{U}_S}{Z_{11}+\frac{(\omega M)^2}{Z_{22}}}\Rightarrow Z_{in}=\frac{\dot{U}_S}{\dot{I}_1}=Z_{11}+\frac{(\omega M)^2}{Z_{22}}$$

$$\dot{I}_2=\frac{j\omega M\dot{U}_S}{(Z_{11}+\frac{(\omega M)^2}{Z_{22}})Z_{22}}\Rightarrow$$

$$=\frac{j\omega M\dot{U}_S}{Z_{11}}\cdot\frac{1}{Z_{22}+\frac{(\omega M)^2}{Z_{11}}}$$

等效电路





记 $Z_L=\frac{(\omega M)^2}{Z_{22}}=\frac{(\omega M)^2}{R_{22}+jX_{22}}=\frac{\omega^2M^2R_{22}}{R_{22}^2+X_{22}^2}-j\frac{\omega^2M^2X_{22}}{R_{22}^2+X_{22}^2}=R_L+jX_L$

- Z_L → 副边对原边的引入阻抗.
- R_L → 引入电阻，恒为正，表示副边回路吸收的功率是靠原边供给的.
- X_L → 引入电抗，负号反映了引入电抗与副边电抗性质相反.

注意：当副边开路时，$Z_{in}=Z_{11}$

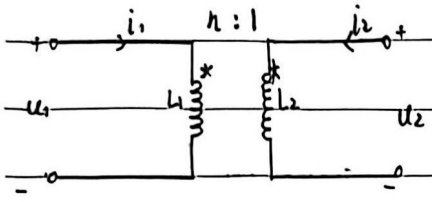
10.5 §. 理想变压器.

1° 三个理想条件.

① 无损耗 → 线圈导线无电阻，做芯子的铁磁材料的磁导率无限大.

② 全耦合 → $k=1 \Rightarrow M=\sqrt{L_1L_2}$

③ 参数无限大 → $L_1, L_2, M \Rightarrow \infty$，但 $\sqrt{\frac{L_1}{L_2}}=\frac{N_1}{N_2}=n$



由无损耗可知.

$\psi_1=\psi_{11}+\psi_{12}=L_1i_1+Mi_2$

$\psi_2=\psi_{20}+\psi_{22}=Mi_1+L_2i_2$.

由全耦合可知

$M=\sqrt{L_1L_2}$

$\phi_1=\phi_2=\phi$

则有 $\frac{\psi_1}{\psi_2}=\frac{\sqrt{L_1}\sqrt{L_1}i_2+\sqrt{L_1}\sqrt{L_2}i_2}{\sqrt{L_1}\sqrt{L_2}i_1+\sqrt{L_2}\sqrt{L_2}i_2}=\frac{\sqrt{L_1}}{\sqrt{L_2}}=\frac{N_1\phi}{N_2\phi}$

$\frac{u_1}{u_2}=\frac{L_1\frac{di_1}{dt}+M\frac{di_2}{dt}}{M\frac{di_1}{dt}+L_2\frac{di_2}{dt}}=\frac{\sqrt{L_1}}{\sqrt{L_2}}$

则有 $\underline{\frac{u_1}{u_2}=\frac{\psi_1}{\psi_2}=\frac{\sqrt{L_1}}{\sqrt{L_2}}=\frac{N_1}{N_2}}$

$L_1, M, L_2 \to \infty$.

$u_1=\frac{d\psi_1}{dt}=L_1\frac{di_1}{dt}+M\frac{di_2}{dt} \Rightarrow i_1=\frac{1}{L_1}\int u_1dt-\frac{M}{L_1}\int\frac{di_2}{dt}dt$.

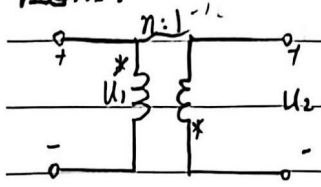
$\Rightarrow i_1=-\sqrt{\frac{L_2}{L_1}}i_2 \Rightarrow \underline{\frac{i_1}{i_2}=-\sqrt{\frac{L_2}{L_1}}}$

2°. 关系. (记 $\sqrt{\frac{L_1}{L_2}}=n$)

① 变压关系

$\frac{u_1}{u_2}=\frac{N_1}{N_2}=\sqrt{\frac{L_1}{L_2}}=n$ （原边若是电压源形成电压，则不能短路）

注意：若.

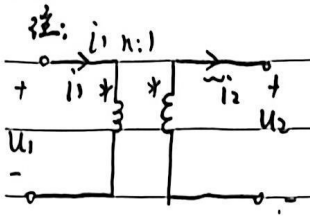


$\frac{u_1}{u_2}=-\frac{N_1}{N_2}=-n$.

①变流关系

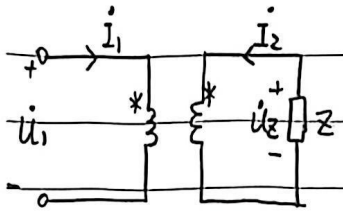
$\frac{i_1}{i_2} = -\frac{1}{n} = -\frac{N_2}{N_1}$ (原边若是电流源形成电流，则不能开路)

注：

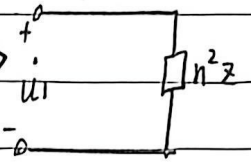


则 $\frac{i_1}{i_2} = \frac{1}{n}$.

②变阻抗关系.



$$\frac{\dot{U}_1}{\dot{I}_1} = \frac{n\dot{U}_2}{(-\frac{1}{n})\dot{I}_2} = n^2(-\frac{\dot{U}_2}{\dot{I}_2}) = n^2 Z \Rightarrow$$



③功率性质

$u_1 = nu_2$.

$i_1 = -\frac{1}{n} i_2$

$P = u_1 i_1 + u_2 i_2 = u_1 i_1 + \frac{1}{n} u_1 \times (-n i_1) = 0$.

表明：a) 理想变压器既不储能，也不耗能，在电路中只起传递信号与能量作用。

b) 理想变压器的特性方程为代数关系，因此也是无记忆元件.

第十一章 电路的频率响应.

§ 11.1 网络函数.

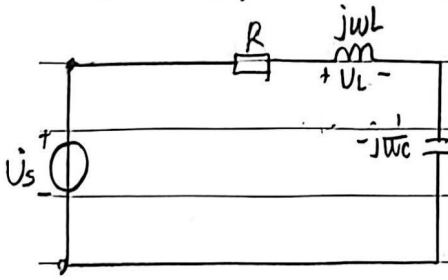
① 网络函数: $H(j\omega)=\frac{响应}{激励}$

② 分类.

a: 驱动点函数: 激励与响应在同一端口. $\begin{cases}\frac{电压}{电流} & 策动点阻抗\\ \frac{电流}{电压} & 策动点导纳.\end{cases}$

b: 转移函数: ~ 不在同一端口 $\begin{cases}\frac{电压}{电流} & 转移阻抗\\ \frac{电流}{电压} & 转移导纳\\ \frac{电压}{电压} & 转移电压比\\ \frac{电流}{电流} & 转移电流比\end{cases}$

§ 11.2 RLC串联电路谐振.



$Z=R+j(\omega L-\frac{1}{\omega C})$

当 $\omega L=\frac{1}{\omega C}$ 时, $Z=R$, 电路达到谐振状态.

$(\omega=\sqrt{\frac{1}{LC}}=2\pi f\rightarrow f=\frac{1}{2\pi\sqrt{LC}})$

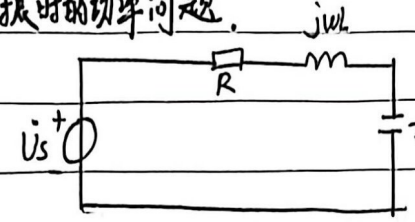
称 ~~$\omega_0=\sqrt{\frac{1}{LC}}$~~ $f=\frac{1}{2\pi\sqrt{LC}}$ 为谐振频率.

$\omega<\omega_0$	$\omega=\omega_0$	$\omega>\omega_0$
容性	纯R.	感性.
$\dot{U}$滞后$\dot{I}$	$\dot{U}$、$\dot{I}$同相.	$\dot{U}$超前$\dot{I}$

② $\dot{U}_L=\frac{\dot{U}_S}{R}j\omega L\Rightarrow|U_L|=\frac{|U_S|}{R}\omega L\Leftrightarrow U_L=\frac{U_S}{1}\cdot\frac{\omega L}{R}$ → 定义为Q品质因数

$=\frac{1}{R}\sqrt{\frac{L}{C}}$.

谐振时的功率问题.

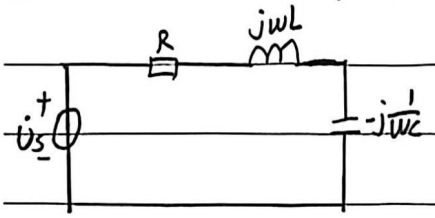


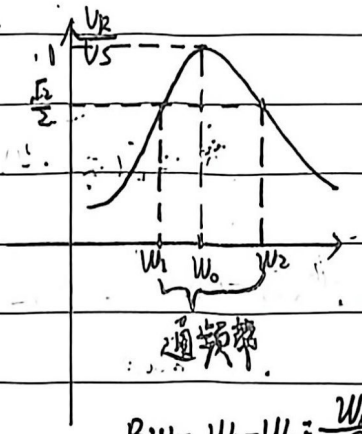
$\tilde{S} = \dot{U}_S(\dot{I})^* = \cancel{U\angle 0 \cdot I\angle 0} = \cancel{UI}$. 只发出有功，不发出无功给外

$$\begin{cases} Q_L = I^2X_L = I^2\omega L & \text{正无功} \\ Q_C = -I^2X_C = -I^2\frac{1}{\omega C} & \text{负无功} \end{cases}$$

$W_{储能} = W_L + W_C = \frac{1}{2}L\left(\frac{U_S}{R}\right)^2 = L\frac{U_S^2}{R} = CQ^2U_S^2$.

§11-3. RLC串联电路的频率响应.



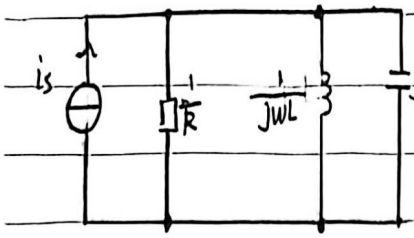


Q越大，越尖锐.

Q越小，胖

$BW_{(带宽)} = \omega_1 - \omega_2 = \frac{\omega_0}{Q}$.

§ 11.4 RLC并联谐振电路.



$$Y=\frac{1}{R}+j\left(\omega C-\frac{1}{\omega L}\right)$$

当 $\omega C=\frac{1}{\omega L}$ 时，即 $\omega_0=\frac{1}{\sqrt{LC}}$，$Y=\frac{1}{R}$ 电路达到电谐振.

$|Y|=\sqrt{\left(\frac{1}{R}\right)^2+\left(\omega C-\frac{1}{\omega L}\right)^2}\geq\frac{1}{R}$ ⇒ 此时 $|Y|$ 最小.

电容、电感两端电压最大.

称 $f_0=\frac{1}{2\pi\sqrt{LC}}$ 为谐振频率.

$\omega<\omega_0$	$\omega=\omega_0$	$\omega>\omega_0$
感性.	$\frac{1}{R}$	容性.
$\dot{U}$ 超前 $\dot{I}$	纯电	$\dot{U}$ 滞后 $\dot{I}$

① 品质因数.

$\dot{U}=\dot{I}_S\cdot R$

$\dot{I}_L=\frac{\dot{I}_S\cdot R}{\omega L}=\frac{R}{\omega_0 L}\dot{I}_S$ ⇒ $I_L=QI_S$ ⇒ $Q=\frac{R}{\omega_0 L}=R\cdot\sqrt{\frac{C}{L}}$

② 功率.

对外

① 只发出有功，不产生无功.

② $W_{储能}=LQ^2I_S^2$ （I_S 有效值）.

第十二章　三相电路.

§12.1　三相电路.

小

三相电源：三个不同相位的电源.

对称三相电源：三个频率相同、幅值相等、相位差依次互差120°的正弦电压源.

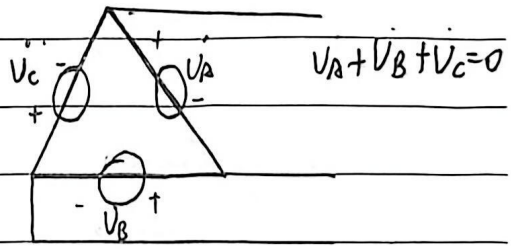
三相负载 { 对称：三个阻抗完全相等
不对称：～　不完全相等.

②、三相电源与三相负载连接方式.

① 三相电源的连接.

a：星形连接(Y接)

b：Δ连接.

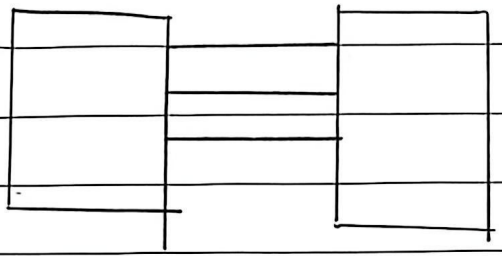


② 三相负载的连接(同上).

③ 三相电源与负载(三相)连接

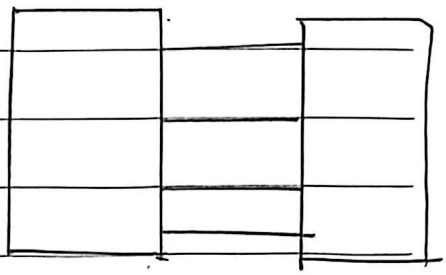
a：三相三线.

Δ/Y —— Δ/Y



b、三相四线

只有Y—Y



§ 12.2 线电压、相电压 & 线电流、相电流关系

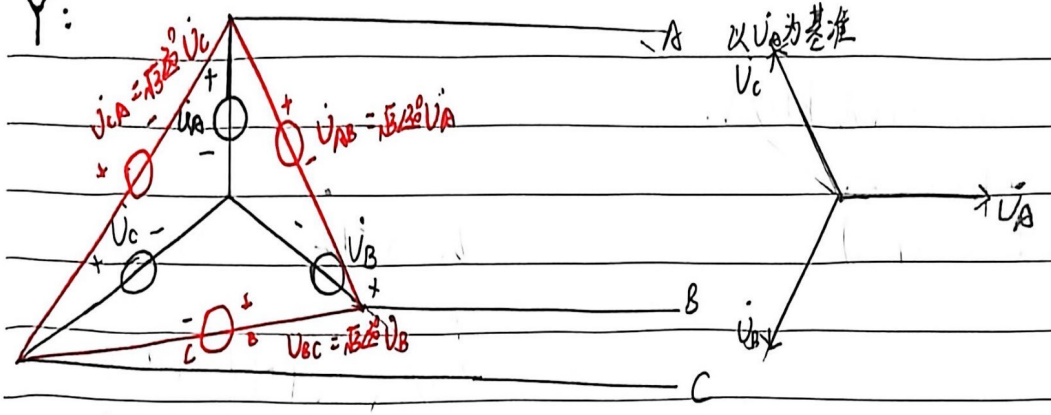
Y

{ 线电压：传输线之间的电压
  线电流：输电线上流过的电流

{ 相电压：每一个二端元件(电压源、阻抗)两端电压
  相电流：每个二端元件上所流过电流。

相电压与线电压：

Y：

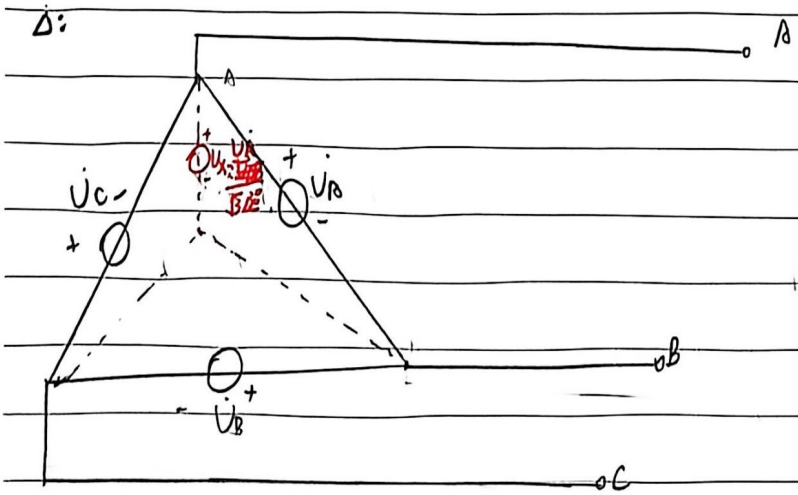


$\dot{U}_{AB}=\dot{U}_A-\dot{U}_B=\sqrt{3}\angle 30^\circ \dot{U}_A$

$\dot{U}_{BC}=\dot{U}_B-\dot{U}_C=\sqrt{3}\angle 30^\circ \dot{U}_B$

$\dot{U}_{CA}=\dot{U}_C-\dot{U}_A=\sqrt{3}\angle 30^\circ \dot{U}_C$

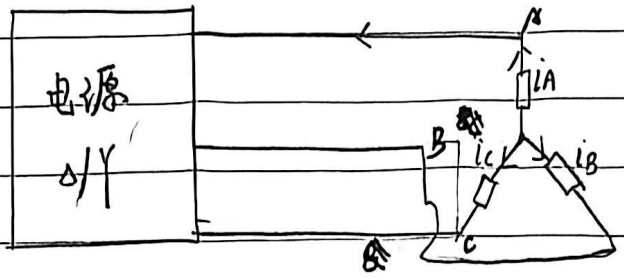
Δ：



{ $\dot{U}_{AB}=\dot{U}_A$
  $\dot{U}_{BC}=\dot{U}_B$
  $\dot{U}_{CA}=\dot{U}_C$

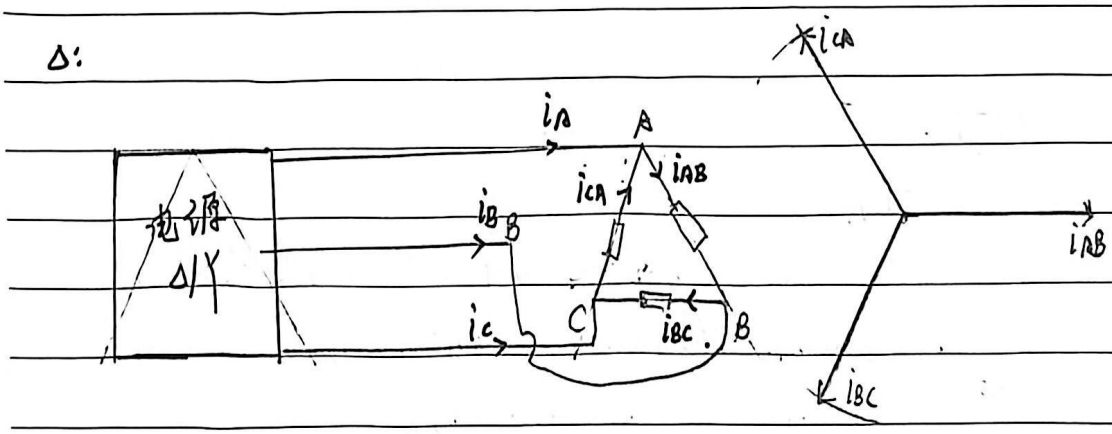
相电流与线电流.

Δ-Y:



$$\begin{cases} \dot{i}_{AB} = \dot{i}_A \\ \dot{i}_{BC} = \dot{i}_B \\ \dot{i}_{CA} = \dot{i}_C \end{cases}$$ 相电流 = 线电流.

Δ:



$$\begin{cases} \dot{I}_A = \sqrt{3}\angle -30^\circ \dot{I}_{AB} \\ \dot{I}_B = \sqrt{3}\angle -30^\circ \dot{I}_{BC} \\ \dot{I}_C = \sqrt{3}\angle -30^\circ \dot{I}_{CA} \end{cases}$$

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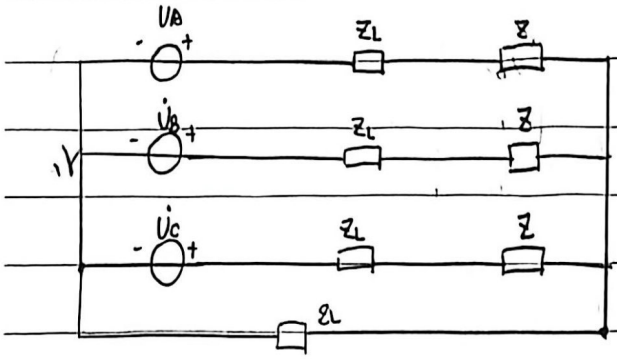
§12.3 对称三相电路计算

Y-Y连接



$$\dot{U}_\varphi\left(\frac{3}{Z_L+Z}+\frac{1}{Z_L}\right)=\frac{\dot{U}_A+\dot{U}_B+\dot{U}_C}{Z_L+Z}=0$$

$\therefore \dot{U}_\varphi=0$

三条回路互不干扰.

$$\dot{I}_A=\frac{\dot{U}_A}{Z_L+Z}$$

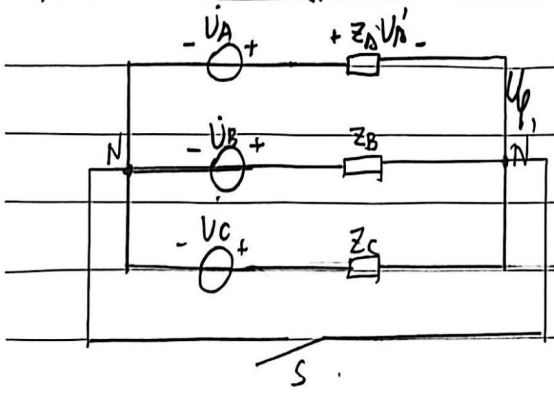
$$\dot{I}_B=\dot{I}_A\angle -120^\circ$$

$$\dot{I}_C=\dot{I}_A\angle 120^\circ.$$

其余类型等效为Y-Y.

§12.4 非对称三相电路计算

图12-6(a)所示电路，若 $Z_A=-j\frac{1}{\omega C}$，而 $Z_B=Z_C=R$，并且 $R=\frac{1}{\omega C}$，则电路是一种测定相序仪器，称为相序指示器(图中电阻R用两个相同的白炽灯代替)。试说明若相电压对称的情况下，当S打开时，如何根据两个白炽灯的亮度确定电源的相序。



$$\dot{U}_\varphi\left(\frac{1}{Z_A}+\frac{1}{Z_B}+\frac{1}{Z_C}\right)=\frac{\dot{U}_A}{Z_A}+\frac{\dot{U}_B}{Z_B}+\frac{\dot{U}_C}{Z_C}$$

$$\dot{U}_\varphi=0.63U\angle 108^\circ$$

$$\dot{U}_A'=\dot{U}_A-\dot{U}_\varphi=$$

$$\dot{U}_B'=\dot{U}_B-\dot{U}_\varphi=1.5U\angle -101.5^\circ$$

$$\dot{U}_C'=\dot{U}_C-\dot{U}_\varphi=0.40U\angle 133.4^\circ$$

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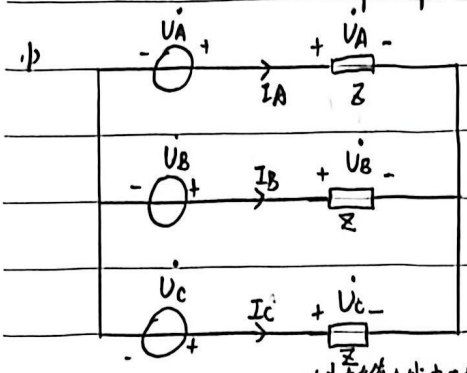
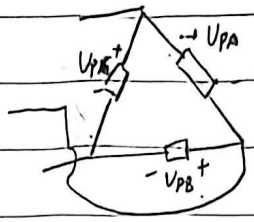
§12.5 对称三相电路的平均功率





P发生负载

U为有交流负载电压相量电流P度.

有功功率: A: $U_A I_A \cos\varphi = U_{PA} I_{PA} \cos\varphi$

B: $U_B I_B \cos\varphi = U_{PB} I_{PB} \cos\varphi$

C: $U_C I_C \cos\varphi = U_{PC} I_{PC} \cos\varphi$

$P_{总} = 3 U_P I_P \cos\varphi$

$= \sqrt{3} \cdot \sqrt{3} U_P I_P \cos\varphi$

$= \sqrt{3} U_L I_L \cos\varphi$

$\begin{cases} U_L = \sqrt{3} U_P \\ I_L = I_P \end{cases}$

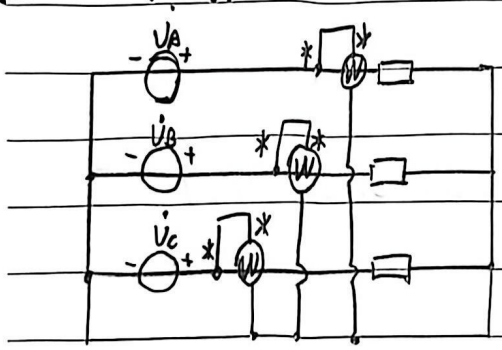
φ 并非为 $\angle(\dot{U}_{AL} - \dot{I}_{AL})$ 而为 $\angle(\dot{U}_{AL} - \dot{I}_{AL})$

无功功率: $Q_{总} = 3 U_P I_P \sin\varphi = \sqrt{3} U_L I_L \sin\varphi$.

视在功率: $S = \sqrt{P_{总}^2 + Q_{总}^2} = \sqrt{3} U_L I_L = 3 U_P I_P$

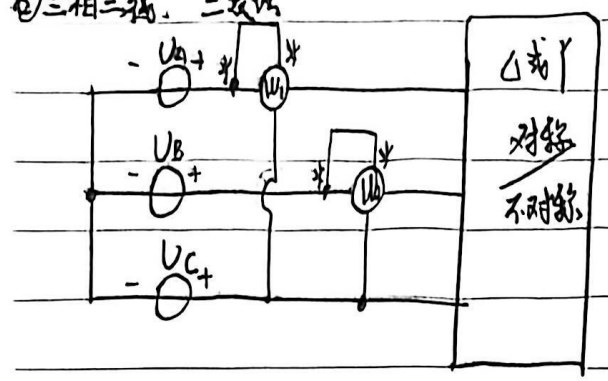
② 三相功率的测量

有功: ① 三相四线.



$W = W_1 + W_2 + W_3$. (对称, 不对称也适用).

① 三相三线，二表法



$$W_1=\frac{\int_0^T u_{AC}i_A dt}{T}$$

$$W_2=\frac{\int_0^T u_{BC}i_B dt}{T}$$

$u_{AC}i_A+u_{BC}i_B=(\varphi_A-\varphi_C)i_A+(\varphi_B-\varphi_C)i_B$

$=\varphi_A i_A+\varphi_B i_B-\varphi_C(i_A+i_B)$

$W_1+W_2=P_{三总}$（对称、不对称都行，$i_A+i_B+i_C=0$）

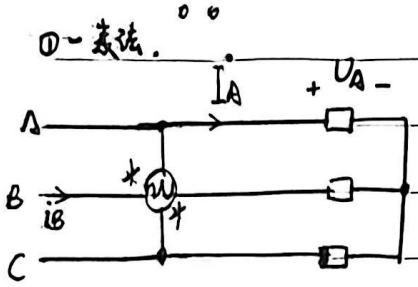
$\because i_A+i_B+i_C=0$

$\therefore u_{AC}i_A+u_{BC}i_B=\varphi_A i_A+\varphi_B i_B+\varphi_C i_C$

$P_{三总}=W_1+W_2$

无功测量（对称三相、三相三线）

① 一表法.



$W=U_{CA}I_B\cos\langle\dot U_{CA},\dot I_B\rangle$

$=U_{CA}I_B\cos(90°-\varphi)$，负载电压超前于负载电流 $\varphi°$

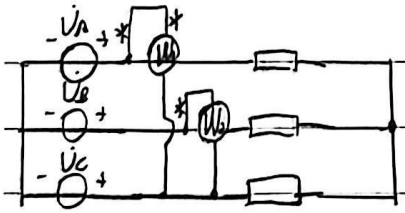
$=U_{CA}I_B\sin\varphi$

$=U_L I_L\sin\varphi$

$Q_{总}=$ ~~三相功~~ $=\sqrt3 W=\sqrt3 U_L I_L\sin\varphi$

（也可以用 $U_{AB}I_C\cos\langle\dot U_{AB},\dot I_C\rangle$，$U_{BC}I_A\cos\langle\dot U_{BC},\dot I_A\rangle$）.

② 二表法.



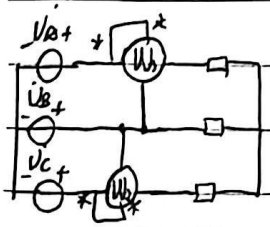
$W_1=U_{AC}I_A\cos\langle\dot U_{AC},\dot I_A\rangle=U_{AC}I_A\cos(\varphi-30°)$

$W_2=U_{BC}I_B\cos(\varphi+30°)$

$$\begin{cases}W_1=U_L I_L\left[\cos\varphi\frac{\sqrt3}{2}+\sin\varphi\cdot\frac12\right]\\ W_2=U_L I_L\left[\cos\varphi\frac{\sqrt3}{2}-\sin\varphi\frac12\right]\end{cases}$$

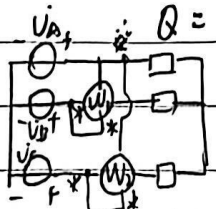
$\therefore W_1-W_2=U_L I_L\sin\varphi$

$Q=\sqrt3(W_1-W_2)\Rightarrow W_A-W_B=\frac{Q}{\sqrt3}$



$Q=\sqrt3(W_2-W_1)$

$W_C-W_A=\frac{Q}{\sqrt3}$



$Q=\sqrt3(W_1-W_2)$

$W_B-W_C=\frac{Q}{\sqrt3}$

$$\begin{cases}\text{共A}\quad(W_B-W_C)\sqrt3=Q_{总}\\ \text{共B}\quad(W_C-W_A)\sqrt3=Q_{总}\\ \text{共C}\quad(W_A-W_B)\sqrt3=Q_{总}\end{cases}$$

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第十三章 非正弦周期电路

§ 13.3 有效值、平均值、平均功率

1) 有效值：→电磁系仪表

$$I=\sqrt{\frac{\int_0^T i^2 dt}{T}}$$

若 $i = I_0 + A_1\cos(\omega t+\varphi_1) + A_2\cos(2\omega t+\varphi_2) + A_3\cos(3\omega t+\varphi_3) + \cdots + A_n\cos(n\omega t+\varphi_n) + \cdots$

则 $I=\sqrt{I_0^2+(\frac{A_1}{\sqrt{2}})^2+(\frac{A_2}{\sqrt{2}})^2+\cdots+(\frac{A_n}{\sqrt{2}})^2+\cdots}$

例：① $i(t) = 5 + 6\sin t + 8\sin 3t + 9\cos 6t$

$$I=\sqrt{5^2+(\frac{6}{\sqrt{2}})^2+(\frac{8}{\sqrt{2}})^2+(\frac{9}{\sqrt{2}})^2}$$

② $i(t) = 5 + 6\sin t + 6\cos t + 8\sin 2t$

$= 5 + 6\sqrt{2}\sin(t+\frac{\pi}{4}) + 8\sin 2t$

$$I=\sqrt{5^2+(\frac{6\sqrt{2}}{\sqrt{2}})^2+(\frac{8}{\sqrt{2}})^2}.$$

2) 平均值：→全波整流仪器

$$I_{av}=\frac{\int_0^T |i|\,dt}{T}.$$

3) 平均功率：

$$P=\frac{\int_0^T ui\,dt}{T}.$$

$u: U_0 + \sqrt{2}U_1\cos(\omega t+\varphi_{u1}) + \sqrt{2}U_2\cos(2\omega t+\varphi_{u2}) + \cdots$

$i: I_0 + \sqrt{2}I_1\cos(\omega t+\varphi_{i1}) + \sqrt{2}I_2\cos(2\omega t+\varphi_{i2}) + \cdots$

$$P=\frac{\int_0^T ui\,dt}{T} = U_0I_0 + U_1I_1\cos(\varphi_{u1}-\varphi_{i1}) + U_2I_2\cos(\varphi_{u2}-\varphi_{i2}) + \cdots$$

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第十四章 线性动态电路的复频域分析.

§. 14.1 拉普拉斯变换的定义.

1. 拉氏变换

$$F(s)=\int_{0_-}^{\infty} f(t)e^{-st}dt. \Leftrightarrow \begin{cases}\mathcal{L}[f(t)]=F(s)\\ \mathcal{L}^{-1}[F(s)]=f(t)\end{cases}$$

§. 14.2 拉氏变换的基本性质.

1. 线性关系.

2. 微分关系.

$$\begin{cases}\mathcal{L}[f(t)]=F(s)\\ \mathcal{L}[f'(t)]=sF(s)-f(0_-)\end{cases}$$

3. 积分性质.

$$\mathcal{L}[f(t)]=F(s)$$

$$\mathcal{L}\left[\int_{0_-}^{t} f(\xi)d\xi\right]=\frac{F(s)}{s}.$$

4. 延迟性质.

$$\mathcal{L}[f(t)]=F(s)$$

$$\mathcal{L}[f(t-t_0)\varepsilon(t-t_0)]=e^{-st_0}F(s)$$

§. 14.3 拉普拉斯反变换的部分式展开.

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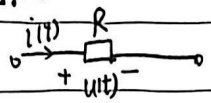
§ 13.4 运算电路.

KCL $\Sigma I(s)=0$

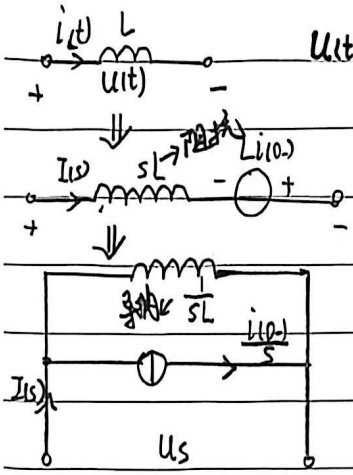
KVL $\Sigma U(s)=0$

电阻.



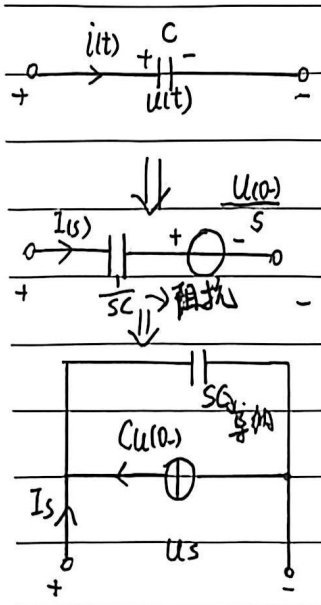
$u(t)=Ri(t)$ ；两边取拉氏变换 $U(s)=RI(s)$

电感.



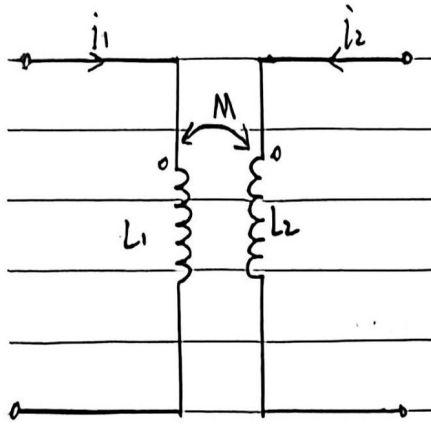
$u(t)=L\frac{di(t)}{dt}$ ；两边取拉氏变换 $\begin{cases} U(s)=sLI(s)-Li(0-) \\ I_s=\frac{1}{sL}U(s)+\frac{i(0-)}{s} \end{cases}$

电容.



$i(t)=C\frac{du_C}{dt}$ ；两边取拉氏变换. $\begin{cases} U(s)=\frac{1}{sC}I(s)+\frac{U} \\ I(s)=sC[U(s)-C[u(0 \end{cases}$

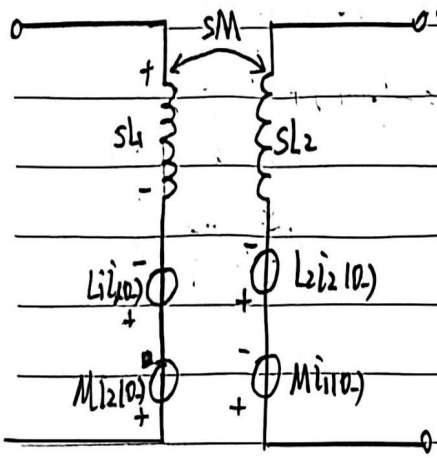
耦合电感：



$$\begin{cases} u_1 = L_1\frac{di_1}{dt} + M\frac{di_2}{dt} \\ u_2 = M\frac{di_1}{dt} + L_2\frac{di_2}{dt} \end{cases}$$

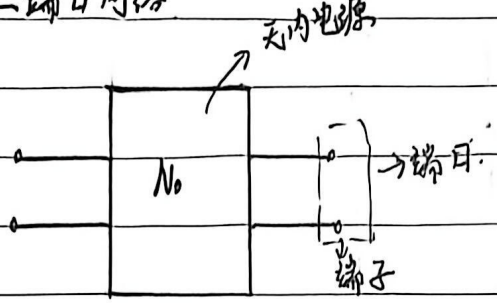
对两边同取拉氏变换，有

$$\begin{cases} U_1(s) = sL_1I_1(s) - L_1i_1(0_-) + sMI_2(s) - Mi_2(0_-) \\ U_2(s) = sL_2I_2(s) - L_2i_2(0_-) + sMI_1(s) - Mi_1(0_-) \end{cases}$$



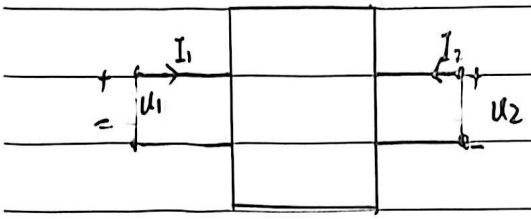
第十六章　二端口网络

§16.1 二端口网络



§16.2 二端口的方程与参数

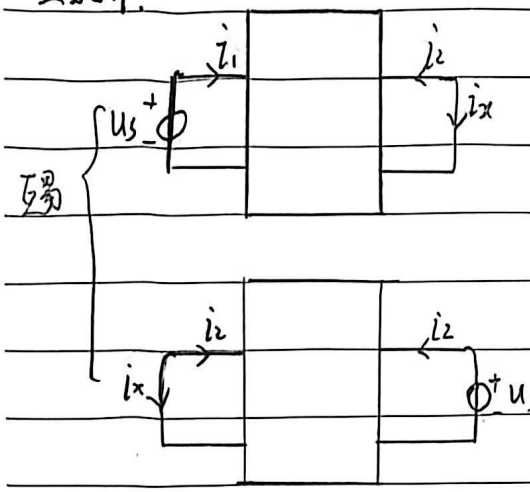
中Y参数矩阵



$$\begin{cases} I_1 = au_1 + bu_2 \\ I_2 = cu_1 + du_2 \end{cases} \Rightarrow \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \underbrace{\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}}_{\text{Y参数矩阵}} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

四Y互易，对称

互易条件：



$$\begin{bmatrix} \dot{I}_1 \\ -\dot{I}_X \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} U_S \\ 0 \end{bmatrix}$$

$$-\dot{I}_X = U_S Y_{21}$$

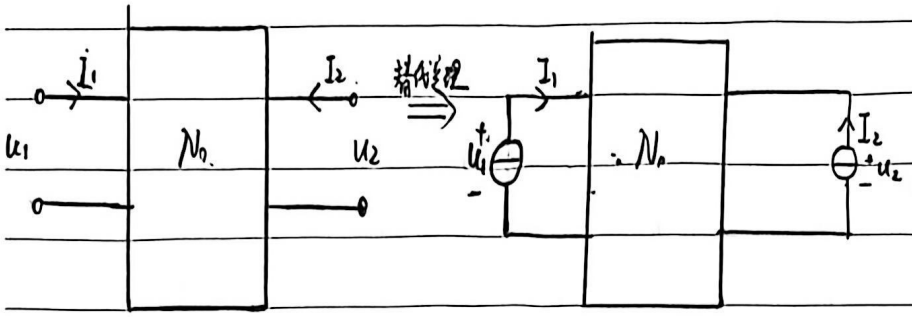
$$\begin{bmatrix} -\dot{I}_X \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} 0 \\ U_S \end{bmatrix}$$

$$-\dot{I}_X = Y_{12} U_S$$

$$\underline{Y_{12} = Y_{21}}$$

对称条件：$Y_{11} = Y_{22}$；$Y_{12} = Y_{21}$.

② Z参数矩阵.



由叠加定理

$$\begin{cases} u_1 = aI_1 + bI_2 \\ u_2 = cI_1 + dI_2 \end{cases} \Rightarrow \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \underbrace{\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}}_{\text{Z参数矩阵}} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Z参数矩阵与[ ]参数矩阵互逆

互易条件：$Z_{12} = Z_{21}$

对称条件 $\begin{cases} Z_{12} = Z_{21} \\ Z_{11} = Z_{22} \end{cases}$

③ T参数矩阵.

$$\begin{cases} \dot{U}_1 = a\dot{U}_2 + b\dot{I}_2 \\ \dot{I}_1 = c\dot{U}_2 + d\dot{I}_2 \end{cases} \Rightarrow \begin{bmatrix} \dot{U}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} a & -b \\ c & -d \end{bmatrix} \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix} = \overbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}^{\text{T参数矩阵}} \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix}$$

互易条件：$AD - BC = 1$

对称条件 $\begin{cases} AD - BC = 1 \\ A = D \end{cases}$

④ H参数矩阵

$$\begin{cases} \dot{U}_1 = a\dot{I}_1 + b\dot{U}_2 \\ \dot{I}_2 = c\dot{I}_1 + d\dot{U}_2 \end{cases} \Rightarrow \begin{bmatrix} \dot{U}_1 \\ \dot{I}_2 \end{bmatrix} = \overbrace{\begin{bmatrix} a & b \\ c & d \end{bmatrix}}^{\text{H参数矩阵}} \begin{bmatrix} \dot{I}_1 \\ \dot{U}_2 \end{bmatrix}$$

互易条件：~~$ad - bc = 1$~~ ~~$a = d$~~ $b = -c$

对称条件 $\begin{cases} ~~a = d~~ \\ ad - bc = 1 \end{cases}$ $b = -c$

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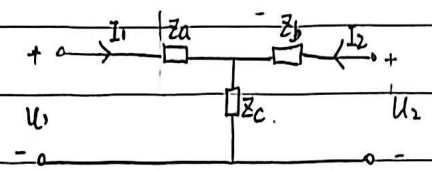
16.3 二端口的等效电路

Z参数矩阵 —— T形等效电路；

Y参数矩阵 —— π形等效电路。

一、互易模型（$Z_{12}=Z_{21}$）

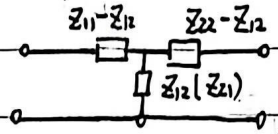


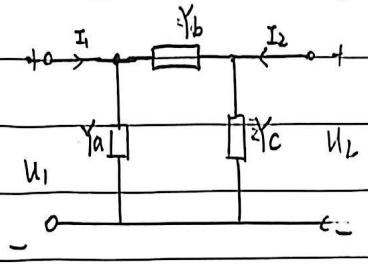
KVL

$$\begin{cases} U_1 = I_1 Z_a + (I_1+I_2)Z_c = (Z_a+Z_c)I_1 + Z_c I_2 \\ U_2 = I_2 Z_b + (I_1+I_2)Z_c = Z_c I_1 + (Z_b+Z_c)I_2 \end{cases}$$

$$\begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

若 $\begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$ ⇒



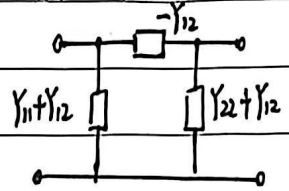


KCL

$$\begin{cases} I_1 = U_1 Y_a + (U_1-U_2)Y_b = U_1(Y_a+Y_b) + (-Y_b)U_2 \\ I_2 = U_2 Y_c + (U_2-U_1)Y_b = U_1(-Y_b) + (Y_c+Y_b)U_2 \end{cases}$$

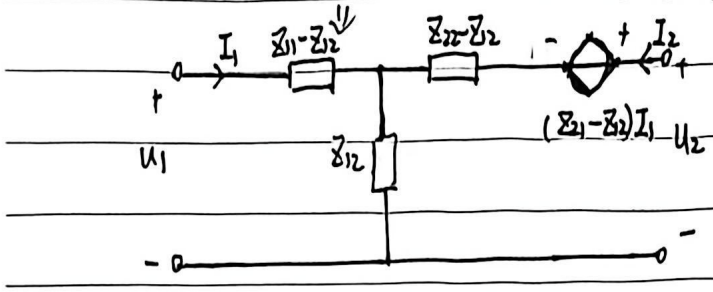
$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_a+Y_b & -Y_b \\ -Y_b & Y_a+Y_b \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$$

若 $\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}$

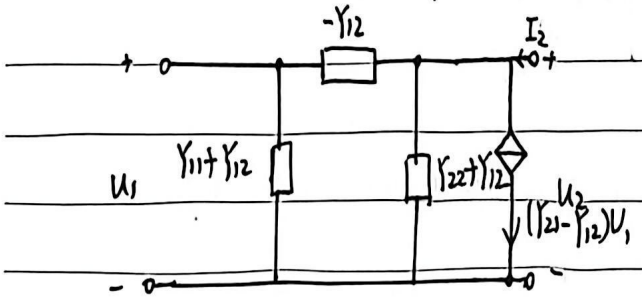


(二) 非互易且非对称(含受控源) $[Z_{12} \neq Z_{21}]$

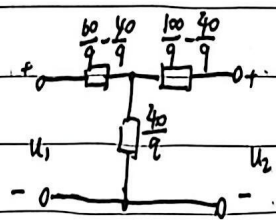
$$\begin{cases} U_1 = Z_{11}I_1 + Z_{12}I_2 \\ U_2 = Z_{21}I_1 + Z_{22}I_2 \end{cases} \Rightarrow \begin{cases} U_1 = Z_{11}I_1 + Z_{12}I_2 \\ U_2 = Z_{12}I_1 + Z_{22}I_2 + (Z_{21} - Z_{12})I_1 \end{cases}$$



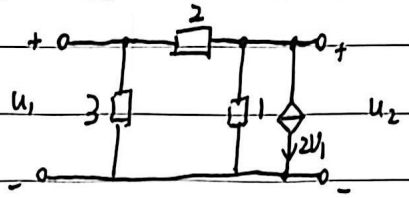
$$\begin{cases} I_1 = U_1Y_{11} + Y_{12}U_2 \\ I_2 = Y_{21}U_1 + Y_{22}U_2 \end{cases} \Rightarrow \begin{cases} I_1 = Y_{11}U_1 + Y_{12}U_2 \\ I_2 = Y_{12}U_1 + Y_{22}U_2 + (Y_{21} - Y_{12})U_1 \end{cases}$$



eg: ① $Z = \begin{bmatrix} \frac{60}{9} & \frac{40}{9} \\ \frac{40}{9} & \frac{100}{9} \end{bmatrix} \Rightarrow$

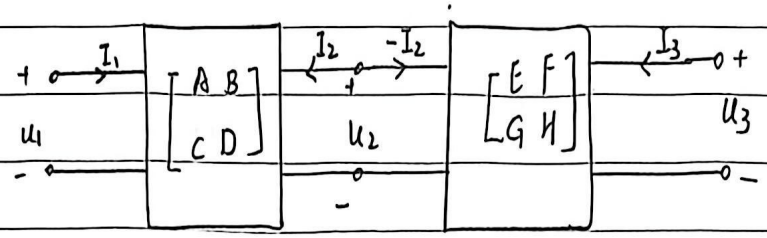


② $Y = \begin{bmatrix} 5 & -2 \\ 0 & 3 \end{bmatrix} \Rightarrow \begin{cases} I_1 = 5U_1 - 2U_2 \\ I_2 = 3U_2 = -2U_2 + 3U_2 + 2U_1 \end{cases}$



§16.5 二端口的连接.

(一) 级联.

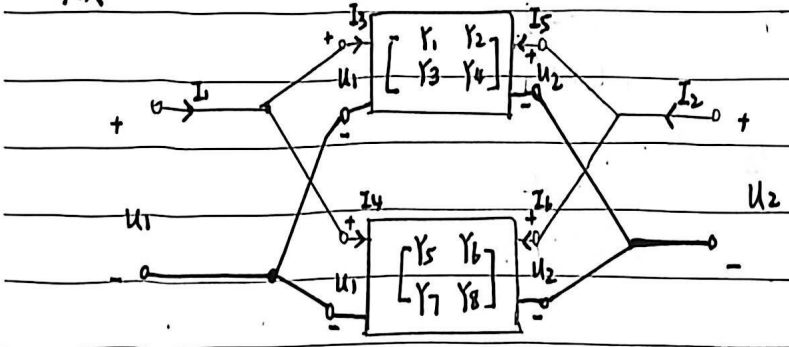


$$\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} U_2 \\ -I_2 \end{bmatrix}$$

$$\begin{bmatrix} U_2 \\ -I_2 \end{bmatrix} = \begin{bmatrix} E & F \\ G & H \end{bmatrix} \begin{bmatrix} U_3 \\ -I_3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix} \begin{bmatrix} U_3 \\ -I_3 \end{bmatrix}$$

(二) 并联



$$\begin{cases} \begin{bmatrix} I_3 \\ I_5 \end{bmatrix} = \begin{bmatrix} Y_1 & Y_2 \\ Y_3 & Y_4 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} \Rightarrow \begin{cases} I_3 = Y_1U_1 + Y_2U_2 \\ I_5 = Y_3U_1 + Y_4U_2 \end{cases} \\ \begin{bmatrix} I_4 \\ I_6 \end{bmatrix} = \begin{bmatrix} Y_5 & Y_6 \\ Y_7 & Y_8 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} \Rightarrow \begin{cases} I_4 = Y_5U_1 + Y_6U_2 \\ I_6 = Y_7U_1 + Y_8U_2 \end{cases} \\ I_1 = I_3 + I_4 \\ I_2 = I_5 + I_6 \end{cases} \Rightarrow \begin{cases} I_1 = (Y_1+Y_5)U_1 + (Y_2+Y_6)U_2 \\ I_2 = (Y_4+Y_3)U_1 + (Y_4+Y_8)U_2 \end{cases}$$

$$\Rightarrow \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_1+Y_3 & Y_2+Y_6 \\ Y_3+Y_7 & Y_4+Y_8 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}$$

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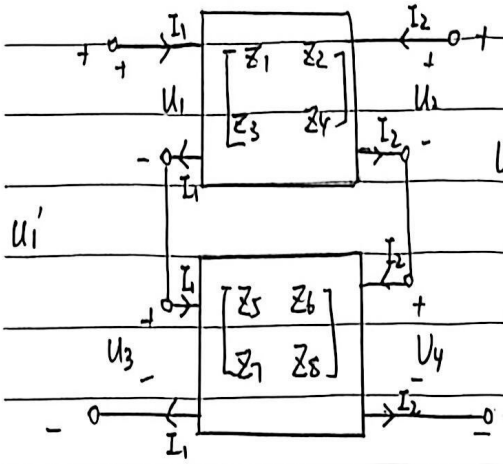
3)串联.



$$\begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} Z_1 & Z_2 \\ Z_3 & Z_4 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \Rightarrow \begin{cases} U_1 = Z_1 I_1 + Z_2 I_2 \\ U_2 = Z_3 I_1 + Z_4 I_2 \end{cases}$$

$$\begin{bmatrix} U_3 \\ U_4 \end{bmatrix} = \begin{bmatrix} Z_5 & Z_6 \\ Z_7 & Z_8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \Rightarrow \begin{cases} U_3 = Z_5 I_1 + Z_6 I_2 \\ U_4 = Z_7 I_1 + Z_8 I_2 \end{cases}$$

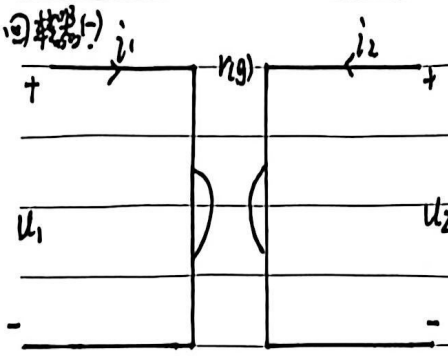
$$U_1' = U_1 + U_3 = I_1(Z_1 + Z_5) + I_2(Z_2 + Z_6)$$

$$U_2' = U_2 + U_4 = I_1(Z_3 + Z_7) + I_2(Z_4 + Z_8)$$

$$\begin{bmatrix} U_1' \\ U_2' \end{bmatrix} = \begin{bmatrix} Z_1+Z_5 & Z_2+Z_6 \\ Z_3+Z_7 & Z_4+Z_8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

§16.6 回转器与负阻抗变换器.

回转器(一):



$$\begin{cases} u_1 = -r i_2 \\ u_2 = r i_1 \end{cases}$$

或.

$$\begin{cases} i_1 = g u_2 \\ i_2 = -g u_1 \end{cases}$$

$r = \frac{1}{g}$.

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 & -r \\ r & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

对偶: $\begin{bmatrix} 0 & g \\ g & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{g} \\ g & 0 \end{bmatrix}$

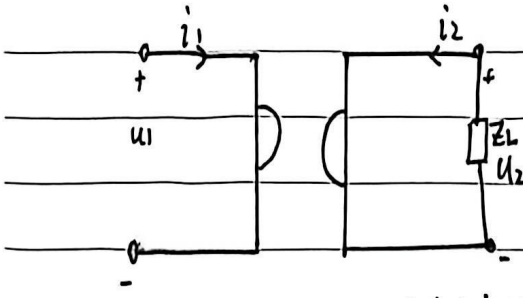
$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{r} \\ -\frac{1}{r} & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$P = u_1 i_1 + u_2 i_2 = -r i_1 i_2 + r i_1 i_2 = 0 \Rightarrow$ 既不消耗功率也不吸收功率.

(二) 回转器的性质

①有把一个端口上的电流"回转"为另一端口上的电压或相反的性质

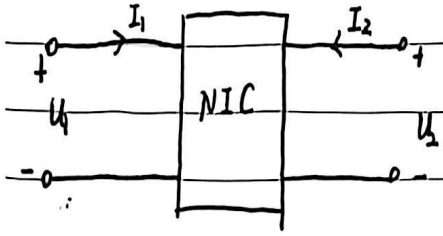
把一个电容回转成一个电感.



$$Z_i = \frac{u_1}{i_1} = \frac{-ri_2}{u_2/r} = \frac{r^2}{Z_L}$$

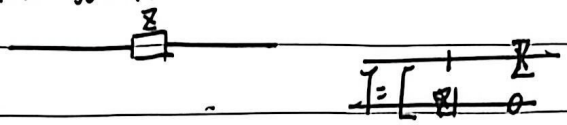
若 $Z_L = \frac{1}{j\omega C} \rightarrow Z_i = r^2 j\omega C$

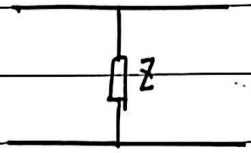
(三) 负阻抗变换器 → 把一个正阻抗变为负阻抗



$$\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -k \end{bmatrix} \begin{bmatrix} U_2 \\ -I_2 \end{bmatrix} \text{ 或 } \begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} -k & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} U_2 \\ -I_2 \end{bmatrix}$$ (k×)

特殊类型T参数矩阵





$$T = \begin{bmatrix} 1 & 0 \\ \frac{1}{Z} & 1 \end{bmatrix}$$